

Hachi: Efficient Lattice-Based Multilinear Polynomial Commitments over Extension Fields

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Abstract. In this work, we present Hachi, a concretely efficient multilinear polynomial commitment scheme that offers succinct proof sizes of $\text{poly}(\ell, \lambda)$ and achieves a “square-root” verifier time complexity of $\tilde{O}(\sqrt{2^\ell \lambda})$ for ℓ -variate polynomials under the Module-SIS assumption. Compared to the current state-of-the-art scheme, Greyhound (CRYPTO 2024), Hachi provides an asymptotic improvement of $\tilde{O}(\lambda)$ in verification time, which translates into a practical 12.5-fold speedup, while maintaining compact proofs of approximately 55 KB.

To improve the verification time, we adopt the sumcheck protocol. Note that the standard sumcheck has efficiency bottlenecks for lattice-based constructions, since lattice operations are usually performed over power-of-two cyclotomic rings $\mathbf{R}_q := \mathbb{Z}_q[X]/(X^d + 1)$. To address this challenge, we provide a novel approach that integrates Greyhound with the ring-switching idea proposed by Huang, Mao and Zhang (ePrint 2025). Surprisingly, under this approach, the verifier does not need to perform any multiplication over $\mathbf{R}_q$, enabling a much faster verification time. This technique could be of independent interest for building lattice-based SNARKs, particularly for achieving faster verification.

As a separate contribution, we introduce a generic reduction that converts polynomial evaluation proofs over extension fields $\mathbb{F}_{q^k}$ (under suitable parameter regimes) into equivalent statements over cyclotomic rings $\mathbf{R}_q$. This reduction is compatible with existing lattice-based polynomial commitment schemes and can be integrated as a modular enhancement to broaden applicability to statements over extension fields.

1 Introduction

Multilinear polynomial commitment scheme (PCS) [KZG10; BBHR18] is a strong cryptographic primitive that allows a prover to commit to a ℓ -dimensional multilinear polynomial $f \in \mathbb{F}^{\leq 1}[X_1, \dots, X_\ell]$ and later prove its evaluation

$$f(x_1, \dots, x_\ell) = y$$

where the evaluation point $(x_1, \dots, x_\ell) \in \mathbb{F}^\ell$ and the image $y \in \mathbb{F}$ are public. Multilinear polynomial commitments are a core building block for succinct non-interactive arguments of knowledge (SNARKs) [WTs+18; Set20; CHM+20; GLS+23; CBBZ23], look-up arguments [STW24] and multi-party computation [BHV+23]. In such applications, it is crucial for the polynomial evaluation proof to be succinct, i.e. polynomial in the number of variables ℓ , and be verified efficiently.

With the rapid advancement of quantum computing, there is increasing interest in quantum-safe PCS constructions. Among these, lattice-based and hash-based approaches have emerged as the leading candidates. Lattice-based PCS are particularly attractive due to their efficient prover runtime and compact proof sizes, both concretely [NS24] and asymptotically [KLNO25]. In contrast, hash-based constructions currently offer significantly faster verification (by two orders of magnitude) compared to lattice-based schemes [ACFY24; ACFY25]. Such a long verification time has become a major bottleneck for lattice-based protocols, such as Greyhound, being widely deployed in practice. This discrepancy raises a natural research question:

Can we design a lattice-based PCS that has fast verification while retaining the compact proof sizes?

In this paper, we answer this question affirmatively.

1.1 Prior Works

Split-and-fold. At the core of state-of-the-art lattice-based succinct proof systems (including polynomial commitments) lies proving knowledge of vectors $\mathbf{s}_1, \dots, \mathbf{s}_r \in \mathbf{R}_q^m$ such that

$$\mathbf{B} \begin{bmatrix} \mathbf{A} & & \\ & \ddots & \\ & & \mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_r \end{bmatrix} = \mathbf{u} \pmod{q} \quad \text{and} \quad \|\mathbf{s}_i\| \leq \beta \quad \text{for } i = 1, \dots, r \quad (1)$$

where $\mathbf{R}_q := \mathbb{Z}_q[X]/(X^d + 1)$ for a power-of-two d , and the matrices $\mathbf{A}, \mathbf{B}$, vector $\mathbf{t}$ and norm bound β are public. The first interactive protocol for this relation was proposed by Baum et al. [BBC+18] which can be summarized as follows. The prover starts by sending the “partial evaluations” $\mathbf{t}_i = \mathbf{A}\mathbf{s}_i$ for $i \in [r]$ (this step is called *split*). The verifier responds by outputting short challenges $c_1, \dots, c_r \in \mathcal{C}$ over the cyclotomic ring $\mathbf{R}_q$. Finally, the prover sends returns the linear combination of witness vectors (this step is called *fold*)

$$\mathbf{z} := c_1 \mathbf{s}_1 + \dots c_r \mathbf{s}_r \in \mathbf{R}_q^m. \quad (2)$$

The verifier then checks three conditions:

$$\mathbf{B} \begin{bmatrix} \mathbf{t}_1 \\ \vdots \\ \mathbf{t}_r \end{bmatrix} = \mathbf{u}, \quad \mathbf{A}\mathbf{z} = \sum_{i=1}^r c_i \mathbf{t}_i \quad \text{and} \quad \|\mathbf{z}\| \text{ is short.} \quad (3)$$

The complexity of the communication of the prover is $r + m$ ring elements, which is strictly sublinear in the original witness length $N := r \cdot m$ in terms of $\mathbf{R}_q$ -elements. By picking parameters appropriately, [BBC+18] obtains “square-root” proofs of size $\tilde{O}(\sqrt{N\lambda})$, where λ is the security parameter.

Multiple follow-up works [BLNS20; AFLN24; CMNW24; KLNO24] managed to reduce the proof size further to polylog(N) ring elements. The key insight lies in making the matrix $\mathbf{A}$ to have a tensor structure similar to the matrix on the left-hand side of Equation (1). This allows the prover, instead of sending the final message $\mathbf{z}$ in the clear, to prove knowledge of $\mathbf{z}$ satisfying the verification equation of (3) in a recursive fashion.

Exact norm proof. Arguably, the hardest part of designing proof systems for relations such as (1) is proving that vectors $\mathbf{s}_i$ are actually short. Indeed, knowledge soundness of the original protocol of [BBC+18] only guarantees extraction of (not necessarily short) $\bar{\mathbf{s}}_1, \dots, \bar{\mathbf{s}}_r \in \mathbf{R}_q^m$ of the form

$$\bar{\mathbf{s}}_i := \frac{\bar{\mathbf{z}}_i}{\bar{c}_i} \quad \text{where both } \bar{\mathbf{z}}_i, \bar{c}_i \text{ are short.}$$

There is a line of research [BLNS20; AL21; ACK21; CLM23; FMN24; AFLN24; KLNO24] which addresses this issue using so-called subtractive sets. A set $\mathcal{C} \subseteq \mathbf{R}_q$ is called subtractive if for any two distinct elements $c, c' \in \mathcal{C}$, inverse of the difference $c - c'$ over $\mathbf{R}_q$ is short. This allows the knowledge extractor to deduce that the extracted $\bar{\mathbf{s}}_i$ are short. This approach has one crucial disadvantage, which is that all subtractive sets are only of polynomial size [AL21]. Consequently, to obtain negligible soundness error, multiple repetitions are needed which affect the efficiency of the scheme significantly.

An alternative approach to proving shortness, which avoids subtractive sets, involves applying random projections to the witness vector $\mathbf{s}$ [BL17; LNS21; LNP22; GHL22; BS23; KLNO25]. Let $\mathbf{s}' \in \mathbb{Z}_q^{md}$ denote the coefficient vector of $\mathbf{s} \in \mathbf{R}_q^m$. Consider a matrix $\mathbf{J} \in \mathbb{Z}^{r \times md}$ with small, independently chosen random entries, where r is a soundness parameter. The Johnson-Lindenstrauss Lemma (and heuristic variants [Ach03; GHL22]) implies that

$$\|\mathbf{J}\mathbf{s}'\| \approx \|\mathbf{s}'\| = \|\mathbf{s}\|.$$

Using this fact, the verifier sends the projection matrix $\mathbf{J}$ to the prover (or in the non-interactive case it is generated from a hash function), who responds with $\mathbf{v} := \mathbf{J}\mathbf{s}' \in \mathbb{Z}^r$. The verifier then checks whether $\mathbf{v}$ is short. For soundness, previous works show that if $\mathbf{s}$ were large (i.e. a prover was malicious), then with probability $\exp(-r)$, $\mathbf{v}$ would also have large norm, and thus result in a contradiction. Hence, $r = O(\lambda)$ is required for soundness.

The prover must also prove that $\mathbf{v}$ is well-formed. As shown in [LNP22; BS23], this reduces to proving that certain r polynomials $u_1, u_2, \dots, u_r \in \mathbf{R}_q$ have zero constant coefficients. One method is to send all $(u_i)_{i \in [r]}$ in the clear for the verifier to check, but this requires sending at least $O(\lambda)$ ring elements, resulting in large proofs. A more efficient method batches the r statements: the verifier sends r random challenges $\alpha_1, \dots, \alpha_r \leftarrow \mathbb{Z}_q$, and the prover returns the linear combination

$$\mathbf{u} := \sum_{i=1}^r \alpha_i \cdot u_i \in \mathbf{R}_q.$$

The verifier checks that the constant coefficient of $\mathbf{u}$ is zero. For soundness, if some u_i had a non-zero constant coefficient, then with probability $1/q$, so would $\mathbf{u}$, leading to a contradiction. Since $1/q$ may not be negligible, this must be repeated $O(\lambda/\log q)$ times to achieve $\exp(-\lambda)$ soundness error.

Concretely, the Johnson-Lindenstrauss projection has shown to be a practical approach to prove norm bounds compared to subtractive sets. The prime example is the LaBRADOR proof system introduced by Beullens and Seiler, which achieves $\approx 60\text{KB}$ proofs for large statements (of size 2^{30}). The main disadvantage of the random projection methodology is the verification time. Indeed, the verifier has to operate on the projection matrix $\mathbf{J}$ of size $O(\lambda \cdot md)$ – linear in witness length (in terms of $\mathbb{Z}_q$ -elements). This makes the LaBRADOR verifier linear time, and thus not suitable for various SNARK-oriented applications. More recently, Klooß et al. [KLNO25] focused on tensor-structured projection matrices $\mathbf{J}$, and managed to reduce the verification time to $\text{polylog}(md)$, at the cost of 40X larger proof sizes compared to LaBRADOR.

Nguyen and Seiler [NS24] proposed a promising hybrid polynomial commitment, called Greyhound. Their evaluation proof intuitively follows the three-round protocol from [BBC+18], but instead of sending the prover message in the clear, the prover commits to $(\mathbf{t}_i)_i, \mathbf{z}$ and later proves that they satisfy the verification equations in (3) via the LaBRADOR proof system. The key advantage of this strategy is that invoking LaBRADOR on a smaller statement yields square-root $\tilde{O}(\lambda\sqrt{N\lambda})$ verification time, while benefiting from the attractive proof sizes of [BS23].

Several recent works employ the sumcheck protocol [LFKN90] to prove exact norm bounds of committed vectors based on lattice assumptions, both in the setting of succinct proofs and folding schemes [KST22]. This strategy was first used by Boneh and Chen in LatticeFold [BC24] to prove that the committed (NTT) coefficients are binary. A drawback of their approach is that the sumcheck protocol is executed over the cyclotomic ring $\mathbf{R}_q$, where each ring element is large, resulting in significantly increased proof sizes. Kuriyama et al. [KLOT25] later expanded this methodology and introduced a new technique to prove exact Euclidean norm bounds using $\mathbf{R}_q$ -based sumchecks. Neo [NS25] departs from $\mathbf{R}_q$ -based sumchecks entirely by mapping all relevant $\mathbf{R}_q$ -relations into $\mathbb{Z}_q$ via rotational (i.e., skew-circulant) matrices. This allows the protocol to run sumcheck directly over $\mathbb{Z}_q$ (or over an extension field $\mathbb{F}_{q^k}$ to obtain negligible soundness error). Concurrently, LatticeFold+ [BC25] proposed a new method to prove range statements. Specifically, to prove that $a \in [0, d)$, the prover commits to the monomial $X^a \in \mathbf{R}_q$ and then proves that the committed ring element is indeed a monomial. LatticeFold+ reduces this (and related statements) to sumcheck claims over $\mathbf{R}_q$. In contrast to [BC24; NS25], the authors observe that these claims do not involve any “full-fledged” $\mathbf{R}_q$ -multiplications and can therefore be naturally reduced to $O(d)$ sumcheck claims over $\mathbb{Z}_q$. The main efficiency bottleneck of this approach is that, for a witness $\vec{s}' \in \mathbb{Z}_q^{md}$ of length md , the prover must commit to md elements of $\mathbf{R}_q$, rather than m as in [BS23; NS24; NS25] (excluding decomposition costs).

1.2 Our Contributions

In this work, we propose Hachi, a concretely efficient multilinear polynomial commitment scheme with succinct $\text{poly}(\ell, \lambda)$ proof size and $\tilde{O}(\sqrt{2^\ell \lambda})$ verifier time for ℓ -variate polynomials based on standard lattice assumptions

Scheme	Multi-linear	Extension fields	Asymptotic		Concrete	
			proof	verifier	proof	verifier
Greyhound [NS24]	✗	✗	$\text{poly}(\ell, \lambda)$	$\tilde{O}(\lambda \cdot \sqrt{2^\ell \lambda})$	53KB	2.8s
Hachi	✓	✓	$\text{poly}(\ell, \lambda)$	$\tilde{O}(\sqrt{2^\ell \lambda})$	55KB	227ms

Fig. 1. Overview of Hachi and comparison with the state-of-the-art lattice-based polynomial commitment scheme Greyhound [NS24]. The asymptotic verifier time is measured in $\mathbb{Z}_q$ -operations. We use the $\tilde{O}$ -notation to hide logarithmic factors. The concrete numbers are taken from evaluating the polynomial with $\ell = 30$ variables.

(and the random oracle model for the Fiat-Shamir transformation). Hence, asymptotically we achieve $\tilde{O}(\lambda)$ improvement in verification time compared to Greyhound [NS24]. In Figure 1 we provide more details on the comparison. Our improvements can be summarized as follows.

Faster verification via ring switching. Our protocol mainly follows the framework of Greyhound [NS24]. However, unlike Greyhound, we move away from the random projection strategy and instead adopt a sumcheck-based approach to prove infinity-norm bounds. Although sumcheck is a well-known protocol, we emphasize that the main challenge for lattice-based constructions is that lattice protocols typically involve arithmetic over large polynomial rings. Directly running the sumcheck protocol over such large rings would be highly inefficient.

To address this challenge and keep the protocol efficient, we propose a novel approach that organically integrates the Greyhound framework with the ring-switching technique introduced by Huang, Mao and Zhang [HMZ25]. This technique lifts statements over $\mathbf{R}_q$ to the polynomial ring $\mathbb{F}_{q^k}[X]$, allowing us to evaluate the equations at a random point $X := \alpha \in \mathbb{F}_{q^k}$ and run the sumcheck protocol over the extension field $\mathbb{F}_{q^k}$. Surprisingly, our verification algorithm does not need to perform any cyclotomic ring operations, even though our protocol is based on lattices. Switching the sumcheck protocol to a smaller extension field is crucial to achieving a total verifier complexity of $\tilde{O}(\sqrt{2^\ell \lambda})$. To the best of our knowledge, other than our protocol, we are not aware of any lattice-based SNARK, PCS, or folding scheme that can eliminate the expensive cyclotomic ring operations for verifiers.

Embedding multilinear evaluation claims inside $\mathbf{R}_q$. We provide a generic transformation from proving ℓ -variate polynomial evaluations over finite fields $\mathbb{F}_{q^k}$ to proving $(\ell - \alpha + \kappa)$ -variate polynomial evaluations over “NTT-unfriendly” cyclotomic rings $\mathbf{R}_q$, where $d = 2^\alpha$ and $k = 2^\kappa$. Thus, we generalize the result of SLAP [AFLN24] in two orthogonal directions: (i) we support polynomials over field extensions (for specific parameter sets), rather than just prime-order fields, and (ii) we extend the strategy from univariate to the multivariate case. Using our methodology, one can already adapt Greyhound [NS24] to prove multilinear polynomial evaluations out-of-the-box. We also provide an optimization if the polynomial f itself is only over $\mathbb{F}_q$ while the evaluation points still lie in $\mathbb{F}_{q^k}$.

Flexibility in picking ring dimension and implications. Both Greyhound and LaBRADOR have certain limitations in picking the ring dimension d . One of the main obstacles is when applying random projection strategy, one has to prove that certain ring elements have constant coefficients equal to zero. As described above, this comes at the cost of the prover sending $O(\lambda/\log q)$ ring elements after each iteration of LaBRADOR. Hence, choosing d (equal to 64 in [BS23; NS24]) as small as possible is essential to achieve small proof sizes. In our construction, since we apply the ring switching idea, the sumcheck protocol is always performed over an extension field, which is independent of the size of d . This gives us more freedom in choosing ring dimension d with smaller impact on the communication size.

In particular, picking larger d has two efficiency benefits: (i) the committing time becomes faster thanks to the optimized NTT-based ring multiplication, and (ii) this allows us to define a challenge space $\mathcal{C}$ of *sparse* ring elements - thus speeding up the ring multiplication for computing the folded witness as in (2).

² Concrete verification time for first round of Hachi.

Implementation. We have developed a prototype implementation of Hachi in Rust. While the implementation is still in its early stages, it is amenable to several optimizations—particularly those leveraging SIMD—that could significantly enhance performance. Even at this early stage, the advantages of our protocol are clearly demonstrated. Specifically, we observe:

- faster verification time. Although the implementation of our verifier does not use advanced optimizations such as SIMD, our verification time is already one order of magnitude faster than Greyhound.³
- faster commitment time. Our commitment step is almost identical to that of Greyhound. However, due to the use of ring switching, we are able to choose larger rings; as a result, our commitment time is 3-5 faster than Greyhound⁴.

1.3 Technical Overview

Denote λ as a security parameter. Let $d := 2^\alpha$ be a power-of-two and $\mathbf{R} := \mathbb{Z}[X]/(X^d + 1)$ be the ring of integers of the $2d$ -th cyclotomic field. Take an odd prime q and define $\mathbf{R}_q := \mathbf{R}/(q)$ and $\delta := \lceil \log q \rceil$. For $k \geq 1$, we denote $\mathbb{F}_{q^k}$ to be a finite field of order q^k . In this overview, we consider base-two gadget matrices $\mathbf{G}_n := \mathbf{I}_n \otimes [1 \ 2 \ 4 \ \dots \ 2^{\delta-1}] \in \mathbf{R}_q^{n \times n\delta}$ for $n \geq 1$. We define the standard inverse function $\mathbf{G}_n^{-1} : \mathbf{R}_q^n \rightarrow \mathbf{R}_q^{n\delta}$, which decomposes each entry w.r.t. base 2. In particular, for any $\mathbf{t} \in \mathbf{R}_q^n$, $\mathbf{G}_n^{-1}(\mathbf{t})$ has binary coefficients and $\mathbf{G}_n \mathbf{G}_n^{-1}(\mathbf{t}) = \mathbf{t}$.

For $i \in \mathbb{Z}_{2d}^\times$, we define the Galois automorphism $\sigma_i : \mathbf{R} \rightarrow \mathbf{R}$ defined as $X \mapsto X^i$, and denote the Galois group of automorphisms as $\text{Aut}(\mathbf{R}) := \{\sigma_i : i \in \mathbb{Z}_{2d}^\times\}$. For a subgroup H of $\text{Aut}(\mathbf{R})$, we denote $\mathbf{R}_q^H := \{x \in \mathbf{R}_q : \forall \sigma \in H, \sigma(x) = x\}$, i.e. the ring of elements of $\mathbf{R}_q$ fixed by all automorphisms in H . Also, we denote the $\mathbf{R}_q$ -trace map $\text{Tr}_H : \mathbf{R}_q \rightarrow \mathbf{R}_q^H$ as $\text{Tr}_H(a) := \sum_{\sigma \in H} \sigma(a) \in \mathbf{R}_q^H$.

Reducing multilinear evaluation claims to relations over $\mathbf{R}_q$. Our first step will be to translate the multilinear evaluation over $\mathbb{F}_{q^k}$, i.e. $f(x_1, \dots, x_\ell) = y$, to an equivalent relation over $\mathbf{R}_q$ (or a related ring). The hardness lies in translating operations over field extensions $\mathbb{F}_{q^k}$, such as addition and multiplication, to equivalent operations over the cyclotomic ring $\mathbf{R}_q$. We proceed in two steps. First, we directly identify the finite fields within $\mathbf{R}_q$.

Lemma 1 (Informal). *Let $q = 5 \pmod{8}$ and $k \geq 1$ be a divisor of $d/2$ and consider the subgroup $H := \langle \sigma_{-1}, \sigma_{4k+1} \rangle$ of $\text{Aut}(\mathbf{R})$. Then, $\mathbf{R}_q^H$ is a subfield of $\mathbf{R}_q$ isomorphic to $\mathbb{F}_{q^k}$.*

This lemma allows us to translate proving inner products over $\mathbb{F}_{q^k}$ into relations directly over the cyclotomic ring $\mathbf{R}_q$ using the $\mathbf{R}_q$ -trace map.

Theorem 1 (Informal). *Let $q = 5 \pmod{8}$, k divide $d/2$ and $H := \langle \sigma_{-1}, \sigma_{4k+1} \rangle \subseteq \text{Aut}(\mathbf{R})$. Then, there exists an efficiently computable bijection $\psi : (\mathbf{R}_q^H)^{d/k} \rightarrow \mathbf{R}_q$ such that for any $\mathbf{a}, \mathbf{b} \in (\mathbf{R}_q^H)^{d/k}$,*

$$\text{Tr}_H(\psi(\mathbf{a}) \cdot \sigma_{-1}(\psi(\mathbf{b}))) = \frac{d}{k} \cdot \langle \mathbf{a}, \mathbf{b} \rangle.$$

We now show how to use these two theoretical results. Let $k = 2^\kappa$ be a divisor of $d/2$ and suppose we want to prove a polynomial evaluation claim over $\mathbf{R}_q^H$ ($\cong \mathbb{F}_{q^k}$ by Lemma 1). Denote the coefficients of polynomial f as $(f_i)_{i \in \{0,1\}^\ell} \in \mathbb{F}_{q^k}$. We can explicitly rewrite the evaluation equation as

$$y = \sum_{i \in \{0,1\}^{\ell-\alpha+\kappa}} \left(x_1^{i_1} \cdot \dots \cdot x_{\ell-\alpha+\kappa}^{i_{\ell-\alpha+\kappa}} \right) \cdot \left(\sum_{j \in \{0,1\}^{\alpha-\kappa}} f_{i \parallel j} \cdot x_{\ell-\alpha+\kappa+1}^{j_1} \cdot \dots \cdot x_\ell^{j_{\alpha-\kappa}} \right). \quad (4)$$

³ Concrete verification times for first round of Hachi, and later composed with Greyhound on a smaller instance.

⁴ When compared with the same implementation.

For each $i \in \{0, 1\}^{\ell-\alpha+\kappa}$ define the ring element

$$F_i := \psi \left((f_{i||j})_{j \in \{0,1\}^{\alpha-\kappa}} \right) \quad \text{and also} \quad v := \psi \left((x_{\ell-\alpha+\kappa+1}^{j_1} \cdots x_{\ell}^{j_{\alpha-\kappa}})_{j \in \{0,1\}^{\alpha-\kappa}} \right).$$

Then, using the fact that $\mathbf{R}_q^H$ is stable under the additively homomorphic map tr_H , and by applying Theorem 1 to Equation (4) we get

$$\frac{d}{k} \cdot y = \text{Tr}_H \left(\underbrace{\sum_{i \in \{0,1\}^{\ell-\alpha+\kappa}} \left(x_1^{i_1} \cdots x_{\ell-\alpha}^{i_{\ell-\alpha}} \right) \cdot F_i \cdot \sigma_{-1}(v)}_{Y :=} \right).$$

Therefore, the prover can send a single ring element $Y \in \mathbf{R}_q$, that allows the verifier to check if the $\mathbf{R}_q$ -trace map of $Y \cdot \sigma_{-1}(v)$ is equal to dy/k . Then, we only need to prove that Y is well-formed, which is now an evaluation proof for the $(\ell - \alpha + \kappa)$ -variate polynomial $F := (F_i)_i$ over $\mathbf{R}_q$.

In certain scenarios, the evaluation point might be defined over the extension field $\mathbb{F}_{q^k}$, while the coefficients of the committed polynomial f are over $\mathbb{F}_q$ (which is the case after the first iteration of our recursive proof system). Can we optimize our transformation in this setting without bluntly assuming that $f \in \mathbb{F}_{q^k}^{\leq 1}[X_1, \dots, X_\ell]$? To this end, we use the similar observation as before that a polynomial evaluation claim can be written as

$$y = \sum_{i \in \{0,1\}^\kappa} \left(x_1^{i_1} \cdots x_\kappa^{i_\kappa} \right) \cdot \underbrace{\left(\sum_{j \in \{0,1\}^{\ell-\kappa}} f_{i||j} \cdot x_{\kappa+1}^{j_1} \cdots x_\ell^{j_{\ell-\kappa}} \right)}_{y_i :=}. \quad (5)$$

Hence, we let the prover send k partial evaluations $y_i \in \mathbb{F}_{q^k}$ to the verifier, who can directly check the equation above⁵. What we have left to prove is that all the y_i are well-formed, or in other words $f_i(x_{\kappa+1}, \dots, x_\ell) = y_i$ where $f_i \in \mathbb{F}_q[X_{\kappa+1}, \dots, X_\ell]$.

Next, denote $\mathbb{F}_{q^k} := \mathbb{F}_q[Z]/\varphi(Z)$, where φ is an irreducible polynomial in $\mathbb{F}_q[Z]$ of degree k . Then, define the $(\ell - \kappa)$ -variate multilinear polynomial $f' \in \mathbb{F}_{q^k}[X_{\kappa+1}, \dots, X_\ell]$:

$$f'(X_{\kappa+1}, \dots, X_\ell) := \sum_{i \in \{0,1\}^\kappa} f_i(X_{\kappa+1}, \dots, X_\ell) \cdot Z^{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}}.$$

Then, it is easy to see that

$$f'(x_{\kappa+1}, \dots, x_\ell) = \sum_{i \in \{0,1\}^\kappa} y_i \cdot Z^{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \in \mathbb{F}_{q^k}$$

which is now a full-fledged polynomial evaluation over $\mathbb{F}_{q^k}$. By applying the generic transformation above, we can reduce the statement into an evaluation claim over $\mathbf{R}_q$ for a $(\ell - \alpha)$ -variate multilinear polynomial (instead of $\ell - \alpha + \kappa$, if done generically).

Polynomial evaluation claim as a quadratic equation. Now, consider proving an evaluation claim of a μ -variate multilinear polynomial f over $\mathbf{R}_q$. We use the following folklore strategy. Suppose $\mu = m + r$ for some $m, r \geq 1$ and let $\mathbf{f} := (f_i)_{i \in \{0,1\}^\mu}$ be the coefficient vector of f . Then, for any

$$f(X_1, \dots, X_\mu) := \sum_{i \in \{0,1\}^r} \sum_{j \in \{0,1\}^m} f_{i||j} \cdot (X_1^{i_1} \cdots X_r^{i_r}) \cdot (X_{r+1}^{j_1} \cdots X_\mu^{j_m})$$

⁵ A simple optimization is for the prover to send only $k - 1$ partial evaluation y_i , excluding $y_{0\dots 0} \in \mathbb{F}_{q^k}$, which can be deterministically derived from the verification equation.

and any evaluation point $(x_1, \dots, x_\mu) \in \mathbf{R}_q^\mu$ we can write:

$$f(x_1, \dots, x_\mu) = \mathbf{b}^\top \begin{bmatrix} \mathbf{a}^\top & & \\ & \ddots & \\ & & \mathbf{a}^\top \end{bmatrix} \begin{bmatrix} \mathbf{f}_{0\dots 0} \\ \vdots \\ \mathbf{f}_{1\dots 1} \end{bmatrix} \quad \text{where } \begin{aligned} \mathbf{b}^\top &:= (x_1^{i_1} \cdot \dots \cdot x_r^{i_r})_{i \in \{0,1\}^r} \in \mathbf{R}_q^{2^r} \\ \mathbf{a}^\top &:= (x_{r+1}^{j_1} \cdot \dots \cdot x_\mu^{j_m})_{j \in \{0,1\}^m} \in \mathbf{R}_q^{2^m} \\ \mathbf{f}_i^\top &:= (f_{i||j})_{j \in \{0,1\}^m} \in \mathbf{R}_q^{2^m} \text{ for } i \in \{0,1\}^r \end{aligned}.$$

Finally, by defining the decomposition $\mathbf{s} := \mathbf{G}_{2^\mu}^{-1}(\mathbf{f}) \in \mathbf{R}_q^{2^{m\delta}}$ and using the mixed product of the tensor product, we deduce that

$$\begin{aligned} f(x_1, \dots, x_\mu) &= \mathbf{b}^\top (\mathbf{a}^\top \otimes \mathbf{I}_{2^r}) \mathbf{f} \\ &= \mathbf{b}^\top (\mathbf{a}^\top \otimes \mathbf{I}_{2^r}) (\mathbf{g}^\top \otimes \mathbf{I}_{2^{m+r}}) \mathbf{s} \\ &= \mathbf{b}^\top (\mathbf{a}^\top (\mathbf{g}^\top \otimes \mathbf{I}_{2^r}) \otimes \mathbf{I}_{2^r}) \mathbf{s}. \end{aligned}$$

Hence, the goal is to prove that the (committed) vector $\mathbf{s}$ is short and satisfies the quadratic equation above, which exactly the relation formulated in Equation (1). To prove knowledge of such $\mathbf{s}$, we closely follow the initial methodology from Greyhound [NS24]. That is, we consider the square-root interactive proof from [BBC+18], but instead of sending the prover messages in the clear, we commit to them. Later, we need to prove knowledge of the prover messages that satisfy all the verification equations (as well as proving knowledge of the commitment opening). All these relations can be phrased as proving knowledge of a short vector $\mathbf{z}$ of dimension $(2^m + 2^r) \cdot \text{poly}(\lambda)$ such that

$$\mathbf{M}\mathbf{z} = \mathbf{w} \quad \text{and} \quad \mathbf{z} \text{ is short.} \quad (6)$$

This is the point where our strategy diverges from [NS24]. Instead of running LaBRADOR [BS23] to prove the relation above, we propose a new sumcheck-based solution.

Ring switching and sumcheck over extension fields. We take inspiration from the ring switching approach proposed recently by Huang, Mao, and Zhang [HMZ25]. To begin with, recall that a multilinear extension $\text{mle}[f] \in \mathbb{F}_{q^k}^{\leq 1}[X_1, \dots, X_\mu]$ of f is defined as

$$\text{mle}[f](\mathbf{x}) := \sum_{i \in \{0,1\}^\mu} f(\mathbf{b}) \cdot \text{eq}(i, \mathbf{x})$$

where $\text{eq}(i, \mathbf{x}) := \prod_{j=1}^\mu ((1 - i_j) \cdot (1 - x_j) + i_j x_j)$ is the equality polynomial.

The first step is to lift Equation (6) from $\mathbf{R}_q$ to $\mathbb{Z}_q[X]$. For the sake of overview, we only focus on the unstructured linear relation and suppose matrix $\mathbf{M}$ has only one row (dealing with multiple rows and proving norm bounds is presented in Section 4.3). Then, we can rewrite it as

$$\sum_k M_k(X) \cdot z_k(X) = w(X) + (X^d + 1) \cdot r(X) \quad \text{for some } r \in \mathbb{Z}_q^{< d-1}[X].$$

Let $\mathbf{z}'$ (resp. $\mathbf{r}'$) be the $\mathbb{Z}_q$ -coefficient vector of $\mathbf{z}$ (resp. r). Then, the prover commits, by using a polynomial commitment scheme, to the multilinear extension $P := \text{mle}[(\mathbf{z}', \mathbf{r}')] \in \mathbb{F}_{q^k}^{\leq 1}[X_1, \dots, X_\mu]$ of $(\mathbf{z}', \mathbf{r}')$ ⁶, before obtaining a challenge $\alpha \leftarrow \mathbb{F}_{q^k}$ from the verifier. The challenge is then used to substitute $X = \alpha$, which simplifies to an inner product claim purely over $\mathbb{F}_{q^k}$. We note that the size of P is much smaller than that of the original committed polynomial f ; hence, we can recursively repeat this process. Similarly as in [HMZ25] this can further get transformed into proving a sumcheck-type relation

$$\sum_{i \in \{0,1\}^\mu} P(i) \cdot Q(i) = V$$

⁶ Here, we implicitly treat a vector $\mathbf{a} \in \mathbb{F}_q^{2^L}$ as an indexing function from $\{0,1\}^L$ to $\mathbb{F}_q$.

for a public polynomial Q over $\mathbb{F}_{q^k}[X_1, \dots, X_\mu]$ and a public value $V \in \mathbb{F}_{q^k}$.

On the other hand, to prove that $\mathbf{z}$ is short, we only need to show that all coordinates of $\mathbf{z}'$ are small. Here we can use a standard range proof over a finite field, since the ring switching has transformed the statements from polynomial rings into a finite field. In contrast, both LaBRADOR and LatticeFold+ demonstrate smallness directly over polynomial rings, which is considerably more complicated and less efficient for the verifier.

Finally, by running the sumcheck protocol [LFKN90], we end up with a polynomial evaluation claim of the form:

$$P(r_1^*, \dots, r_\mu^*) \cdot Q(r_1^*, \dots, r_\mu^*) = y^*,$$

where the evaluation point $(r_1^*, \dots, r_\mu^*)$ and the image y^* are public. Hence, by sending the evaluation $P(r_1^*, \dots, r_\mu^*)$ to the verifier, they can directly check the equation above. What is left for the prover to do now is to prove the evaluation claim for the polynomial P , and thus the protocol can be repeated in a further way to obtain smaller proof sizes.

Committing to P and avoiding re-decomposition. A naive approach to commit to the multilinear polynomial $P = \text{mle}[(\mathbf{z}', \mathbf{r}')]$ is to first decompose the coefficients of P (and thus the committed witness gets longer), and then apply an Ajtai-style [Ajt96] commitment scheme. A more natural and efficient way is to directly commit to $(\mathbf{z}', \mathbf{r}')$, as we know that the coefficients of $\mathbf{z}'$ are already small and thus no decomposition for $\mathbf{z}'$ is needed⁷. But this directly leads to a question of how to directly prove $\text{mle}[(\mathbf{z}', \mathbf{r}')](\mathbf{x}_1, \dots, \mathbf{x}_\mu) = y'$, when having only committed to $(\mathbf{z}', \mathbf{r}')$. We solve this issue by observing a very simple homomorphic property of the eq polynomials. Namely, for any vectors i, j and $\mathbf{x}_0, \mathbf{x}_1$ we have

$$eq(i, \mathbf{x}_0) \cdot eq(j, \mathbf{x}_1) = eq(i \parallel j, \mathbf{x}_0 \parallel \mathbf{x}_1).$$

Then, by definition of the multilinear extension and for $\mu = m + r$, we can write

$$\begin{aligned} \text{mle}[(\mathbf{z}', \mathbf{r}')](\mathbf{x}_0 \parallel \mathbf{x}_1) &= \sum_{i \in \{0,1\}^r} \sum_{j \in \{0,1\}^m} (\mathbf{z}', \mathbf{r}')_{i \parallel j} \cdot eq(i \parallel j, \mathbf{x}_0 \parallel \mathbf{x}_1) \\ &= \sum_{i \in \{0,1\}^r} eq(i, \mathbf{x}_0) \cdot \left(\sum_{j \in \{0,1\}^m} (\mathbf{z}', \mathbf{r}')_{i \parallel j} \cdot eq(j, \mathbf{x}_1) \right) \end{aligned}$$

which is exactly of the same form as (4). Therefore, we can recursively run our protocol to prove evaluation of P (with minor modifications) without explicitly committing to its coefficients.

2 Preliminaries

Notation. Let $q = 5 \pmod{8}$ be an odd prime. Denote $\mathbb{Z}_q := \{0, 1, \dots, q-1\}$ to be the ring of integers of order q and $\mathbb{F}_{q^k}$ be the Galois field of dimension k and size q^k . In this paper, we will use $\mathbb{Z}_q$ and $\mathbb{F}_q$ interchangeably. For $n \in \mathbb{N}$, we define $[n] := \{1, 2, \dots, n\}$. Let λ to be the security parameter. We write $O_\lambda(T)$ to denote $T \cdot \text{poly}(\lambda)$. For a probability distribution $\mathcal{X}$ (resp. finite set $\mathcal{X}$), $x \leftarrow \mathcal{X}$ means that x is sampled from $\mathcal{X}$ (resp. x is chosen uniformly at random from the set $\mathcal{X}$). We write negl to denote an unspecified negligible function.

We will mostly use letters i, j for indexing. For $n \in \mathbb{N}$, we write $[n] := \{1, 2, \dots, n\}$. When we denote $i \in \{0, 1\}^k$, we treat i as a binary vector $(i_1, \dots, i_k)$ of length k . For presentation purposes, when we say $i \in [2^k]$, we may also mean its binary encoding $i \in \{0, 1\}^k$. This will be useful when describing the sumcheck-based norm protocol (cf. Section 4.3).

⁷ Nevertheless, we would need to decompose the coefficients of $\mathbf{r}'$.

2.1 Cyclotomic Rings

For a power of two $d := 2^\alpha$ and prime q , denote $\mathbf{R}$ and $\mathbf{R}_q$ respectively to be the rings $\mathbb{Z}[X]/(X^d + 1)$ and $\mathbb{Z}_q[X]/(X^d + 1)$. Lower-case letters denote elements in $\mathbf{R}$ or $\mathbf{R}_q$ and bold lower-case (resp. upper-case) letters represent column vectors (resp. matrices) with coefficients in $\mathbf{R}$ or $\mathbf{R}_q$. For $y = \sum_{i=0}^{d-1} y_i \cdot X^i \in \mathbf{R}_q$, we write $\text{cf}(y) := (y_0, \dots, y_{d-1}) \in \mathbb{F}_q^d$ to denote the coefficient vector of y . We naturally extend this notation to vectors.

We consider base- b gadget matrices $\mathbf{G}_{b,n} := \mathbf{I}_n \otimes [1 \ b \ b^2 \ \dots \ b^{\delta-1}] \in \mathbf{R}_q^{n \times n\delta}$ for $n \geq 1$ and $\delta = \lceil \log_b q \rceil$. We define the inverse function $\mathbf{G}_{b,n}^{-1} : \mathbf{R}_q^n \rightarrow \mathbf{R}_q^{n\delta}$, which decomposes each entry w.r.t. base b . In particular, for any $\mathbf{t} \in \mathbf{R}_q^n$, $\mathbf{G}_{b,n}^{-1}(\mathbf{t})$ has coefficients between $\lceil -b/2 \rceil$ and $\lceil b/2 \rceil - 1$ and $\mathbf{G}_{b,n} \mathbf{G}_{b,n}^{-1}(\mathbf{t}) = \mathbf{t}$. We will omit the subscript b if the base is known and fixed from the context.

Galois Automorphisms and trace map. For $i \in \mathbb{Z}_{2d}^\times$, we define the Galois automorphism $\sigma_i : \mathbf{R} \rightarrow \mathbf{R}$ defined as $X \mapsto X^i$, and denote the Galois group of automorphisms as $\text{Aut}(\mathbf{R}) := \{\sigma_i : i \in \mathbb{Z}_{2d}^\times\}$. For a subgroup H of $\text{Aut}(\mathbf{R})$, we denote $\mathbf{R}_q^H := \{x \in \mathbf{R}_q : \forall \sigma \in H, \sigma(x) = x\}$, i.e. the ring of elements of $\mathbf{R}_q$ fixed by all automorphisms in H . Also, we define the $\mathbf{R}_q$ -trace map $\text{Tr}_H : \mathbf{R}_q \rightarrow \mathbf{R}_q^H$ as $\text{Tr}_H(a) := \sum_{\sigma \in H} \sigma(a) \in \mathbf{R}_q^H$.

Norms. We define $r' = r \bmod^\pm q$ to be the unique element r' in the range $-\frac{q-1}{2} \leq r' \leq \frac{q-1}{2}$ such that $r' = r \bmod q$. We also denote $r' = r \bmod^+ q$ to be the unique element r' in the range $0 \leq r' < q$ such that $r' = r \bmod q$. When the exact representation is not important, we simply write $r \bmod q$. For an element $w \in \mathbb{Z}_q$, we write $\|w\|_\infty$ to mean $|w \bmod^\pm q|$. Define the ℓ_∞ and ℓ_p norms for $w = w_0 + w_1 X + \dots + w_{d-1} X^{d-1} \in \mathbf{R}$ as $\|w\|_\infty = \max_j \|w_j\|_\infty$, $\|w\|_p = \sqrt[p]{\|w_0\|_\infty^p + \dots + \|w_{d-1}\|_\infty^p}$. If $\mathbf{w} = (\mathbf{w}_1, \dots, \mathbf{w}_m) \in \mathbf{R}^k$, then $\|\mathbf{w}\|_\infty = \max_j \|\mathbf{w}_j\|_\infty$, $\|\mathbf{w}\|_p = \sqrt[p]{\|\mathbf{w}_1\|_p^p + \dots + \|\mathbf{w}_k\|_p^p}$. By default, $\|\mathbf{w}\| := \|\mathbf{w}\|_2$. For $\beta \geq 1$, we define S_β to be the set of ring elements in $\mathbf{R}$ with coefficients between $\lceil -\beta/2 \rceil$ and $\lceil \beta/2 \rceil - 1$.

In this work, we will use the well-known inequality of [Mic07] to derive norm bounds.

Lemma 2. For any $f, g \in \mathbf{R}$, $\|fg\|_\infty \leq \|f\|_1 \cdot \|g\|_\infty$.

Short elements are invertible. We recall the main result by Lyubashevsky and Seiler [LS18] which says that short polynomials over $\mathbf{R}_q$ are invertible over $\mathbf{R}_q$.

Lemma 3 ([LS18]). Let $q \equiv 5 \pmod{8}$ be a prime. Then, any $f \in \mathbf{R}_q$ which satisfies $0 < \|f\|_\infty < \frac{1}{\sqrt{2}} q^{1/2}$ or $0 < \|f\| < q^{1/2}$ has an inverse in $\mathbf{R}_q$.

The set of invertible elements of $\mathbf{R}_q$ is denoted by $\mathbf{R}_q^\times$.

Module-SIS Assumption. We recall the Module-SIS hardness assumption [LS15] adapted to the infinity norm setting.

Definition 1 (Module-SIS). Let $q = q(\lambda)$, $n = n(\lambda)$, $m = m(\lambda)$, $d = d(\lambda)$ and $\beta = \beta(\lambda)$. We say that the $\text{MSIS}_{q,d,n,m,\beta}$ assumption holds if for any PPT algorithm $\mathcal{A}$, the following holds:

$$\Pr [\mathbf{A}\mathbf{z} = \mathbf{0} \wedge 0 < \|\mathbf{z}\|_\infty \leq \beta \mid \mathbf{A} \leftarrow \mathbf{R}_q^{n \times m}, \mathbf{z} \leftarrow \mathcal{A}(\mathbf{A})] = \text{negl}(\lambda)$$

2.2 Multilinear Extensions over Rings

Definition 2 (Multilinear extension over rings). Let R be an arbitrary ring with zero 0 and identity 1. Given a function $f : \{0, 1\}^\mu \rightarrow R$, we define the multilinear extension $\text{mle}[f] \in R^{\leq 1}[X_1, \dots, X_\mu]$ of f as

$$\text{mle}[f](\mathbf{x}) := \sum_{i \in \{0,1\}^\mu} f(i) \cdot \text{eq}(i, \mathbf{x})$$

where $\text{eq}(i, \mathbf{x}) := \prod_{j=1}^\mu ((1 - i_j) \cdot (1 - x_j) + i_j x_j)$.

By definition, $\text{mle}[f](\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in \{0, 1\}^\mu$.

For a vector $\mathbf{f} := (f_i)_{i \in \{0,1\}^\mu} \in R^{2^\mu}$, we denote $\tilde{\mathbf{f}}$ as the multilinear extension of the function $f : \{0, 1\}^\mu \rightarrow R$ defined by $i \mapsto f_i$.

2.3 Coordinate-Wise Special Soundness

Let $R \subseteq \{0,1\}^* \times \{0,1\}^* \times \{0,1\}^*$ be a ternary relation. For a triple $(\text{pp}, x, w) \in R$, we call pp the public parameters, x is a statement and w is a witness for x w.r.t. pp ⁸. We denote $R(\text{pp}, x) = \{w : R(\text{pp}, x, w) = 1\}$.

An interactive proof system $\Pi = (\mathcal{S}, \mathcal{P}, \mathcal{V})$ for relation R consists of three PPT algorithms: the setup algorithm $\mathcal{S}$, prover $\mathcal{P}$, and verifier $\mathcal{V}$. The latter two are interactive and stateful. Further, we say a protocol is *public coin* if the verifier's challenges are chosen uniformly at random independently of the prover's messages.

We recall the notion of coordinate-wise special soundness from [FMN24]. Let S be a set and $\ell, k \in \mathbb{N}$. Namely, take two vectors $\mathbf{x} := (x_1, \dots, x_\ell), \mathbf{y} := (y_1, \dots, y_\ell) \in S^\ell$. Then, we define the following relation " $\equiv_i$ " for fixed $i \in [\ell]$ as:

$$\mathbf{x} \equiv_i \mathbf{y} \iff x_i \neq y_i \wedge \forall j \in [\ell] \setminus \{i\}, x_j = y_j.$$

That is, vectors $\mathbf{x}$ and $\mathbf{y}$ have the same values in all coordinates apart from the i -th one. For $\ell = 1$, the relations boil down to checking whether two elements are distinct. Further, we can define the set

$$\text{SS}(S, \ell, k) := \left\{ \{ \mathbf{x}_1, \dots, \mathbf{x}_K \} \subseteq (S^\ell)^K : \begin{array}{l} \exists e \in [K], \forall i \in [\ell], \\ \exists J = \{j_1, \dots, j_{k-1}\} \subseteq [K] \setminus \{e\}, \\ \forall j \in J, \mathbf{x}_e \equiv_i \mathbf{x}_{j_i} \end{array} \right\},$$

where $K := \ell(k-1) + 1$. In other words, there is a *central* vector $\mathbf{x}_e \in X$ such that for each coordinate of $\mathbf{x}_e$, there are $k-1$ other vectors in X that differ from $\mathbf{x}_e$ only in that coordinate. In other words, for each coordinate, there are k vectors in X that differ from each other only in that coordinate, and $\mathbf{x}_e$ is always one of them.

We are ready to define the notion of coordinate-wise special soundness.

Definition 3 (CWSS for Multi-Round Protocols). *Let Π be public-coin $(2\mu+1)$ -round interactive proof system for relation R , where in the $2i$ -th round, for $i \in [\mu]$, the verifier picks a uniformly random challenge from $S_i^{\ell_i}$. A tree of transcripts is a set of $K := \prod_{i=1}^{\mu} (\ell_i(k_i-1) + 1)$ arranged in the following tree structure. The nodes in the tree correspond to the prover's messages and the edges correspond to the verifier's challenges. Each node at depth i has exactly $\ell_i(k_i-1) + 1$ children corresponding to $\ell_i(k_i-1) + 1$ distinct challenges which, as a set of vectors, lie in $\text{SS}(S_i, \ell_i, k_i)$. Every transcript corresponds to exactly one path from the root to a leaf node.*

We say that Π is $(\ell_1, \dots, \ell_\mu)$ -coordinate-wise $(k_1, \dots, k_\mu)$ -special sound for the relation R if there is a polynomial time algorithm that given public parameters pp , statement x and the tree of transcripts, outputs a witness $w \in R(\text{pp}, x)$. If $\ell_1 = \dots = \ell_\mu = 1$ then we say the protocol is $(k_1, \dots, k_\mu)$ -special sound.

We recall the main result of [FMN24, Section 7] which translates CWSS to knowledge soundness.

Lemma 4. *Let Π be a $(\ell_1, \dots, \ell_\mu)$ -coordinate-wise $(k_1, \dots, k_\mu)$ -special sound interactive proof system as in Definition 3 and assume $K := \prod_{i=1}^{\mu} (\ell_i(k_i-1) + 1) = \text{poly}(\lambda)$. Then, it is knowledge sound with knowledge error*

$$\sum_{i=1}^{\mu} \frac{\ell_i \cdot k_i}{|S_i|^{\ell_i}}.$$

A similar result also holds in the random oracle model if Π is turned non-interactive via Fiat-Shamir transformation (see [FMN24, Section 8]). In this work, we will aim to show that our protocol satisfies coordinate-wise special soundness.

⁸ We omit pp and simply write $R(x, w)$ if the public parameters are not relevant to the relation.

3 Evaluation Claims over Extension Fields as Relations over $\mathbf{R}_q$

In this section, we provide a generic transformation from proving ℓ -variate polynomial evaluations over finite fields $\mathbb{F}_{q^k}$ to proving $(\ell - \alpha + \kappa)$ -variate polynomial evaluations over the cyclotomic ring $\mathbf{R}_q$. Next, we improve the result if the committed polynomial is only over the base field $\mathbb{F}_q$, instead of $\mathbb{F}_{q^k}$. At the core of our transformation is the process of embedding vectors of extension fields $\mathbb{F}_{q^k}$ inside ring elements $\mathbf{R}_q$ and translating inner products over $\mathbb{F}_{q^k}$ as “inner-product-related” statements over $\mathbf{R}_q$. In Figure 2 we provide a summary table of how much the $\mathbb{F}_{q^k}$ -to- $\mathbf{R}_q$ transformations presented in this section cost in terms of prover communication.

3.1 Generic Transformation over $\mathbb{F}_{q^k}$

Finding subfields in $\mathbf{R}_q$. We introduce a lemma that explicitly describe finite field extensions within $\mathbf{R}_q$.

Lemma 5 (Subfields of $\mathbf{R}_q$). *Let $q = 5 \pmod{8}$ and $k \geq 1$ be a divisor of $d/2$ and $H := \langle \sigma_{-1}, \sigma_{4k+1} \rangle \subseteq \text{Aut}(\mathbf{R})$. Then, $\mathbf{R}_q^H$ is a subfield of $\mathbf{R}_q$ isomorphic to $\mathbb{F}_{q^k}$.*

Proof. By definition, $\mathbf{R}_q^H$ is a subset of the set of all ring elements in $\mathbf{R}_q$ stable under the σ_{-1} automorphism, which itself is a field [LNP22, Lemma 2.6]. Using the homomorphic properties of Galois automorphisms, the set $\mathbf{R}_q^H$ is closed under both addition and multiplication. Finally, the size of $\mathbf{R}_q^H$ is q^k since each element of $\mathbf{R}_q^H$ is of the form

$$a := a_0 + \sum_{j=1}^{k-1} a_{k-j} \cdot \left(X^{\frac{d}{2k} \cdot (k-j)} - X^{\frac{d}{2k} \cdot (k+j)} \right) \quad (7)$$

with k degrees of freedom: $a_0, a_1, \dots, a_{k-1} \in \mathbb{Z}_q$. □

Inner products. Our main theorem of this section generalizes [LNP22], and reduces proving inner products over the subfield into proving statements over $\mathbf{R}_q$ related to the trace map. Informally, it introduces a bijection ψ which allows packing d/k elements of field $\mathbb{F}_{q^k}$ into one single $\mathbf{R}_q$ element. The fact that it is an invertible function is crucial in arguing knowledge soundness.

Theorem 2 (Inner product as output of the trace). *Let k divide $d/2$ and $H := \langle \sigma_{-1}, \sigma_{4k+1} \rangle \subseteq \text{Aut}(\mathbf{R})$. Consider a map*

$$\begin{aligned} \psi : (\mathbf{R}_q^H)^{d/k} &\rightarrow \mathbf{R}_q, \\ (a_0, a_1, \dots, a_{d/k-1}) &\mapsto \left(\sum_{i=0}^{d/2k-1} a_i \cdot X^i \right) + X^{d/2} \cdot \left(\sum_{i=0}^{d/2k-1} a_{d/2k+i} \cdot X^i \right). \end{aligned} \quad (8)$$

Then, ψ is a bijection. Moreover, for any $\mathbf{a}, \mathbf{b} \in (\mathbf{R}_q^H)^{d/k}$,

$$\text{Tr}_H(\psi(\mathbf{a}) \cdot \sigma_{-1}(\psi(\mathbf{b}))) = \frac{d}{k} \cdot \langle \mathbf{a}, \mathbf{b} \rangle.$$

Proof. To prove the first part, we only need to show that ψ is injective since the domain and co-domain size are the same. Indeed, suppose that for some $\mathbf{a} = (a_0, \dots, a_{d/k-1})$ we have $\psi(\mathbf{a}) = 0$ (the statement holds by the linearly homomorphic property of ψ). We prove that $\mathbf{a}$ is the zero-vector.

First, we note that $\psi(\mathbf{a})$ can be expanded as

$$\begin{aligned} \sum_{i=0}^{\frac{d}{2k}-1} X^i a_{i,0} + X^{\frac{d}{2}+i} a_{d/2k+i,0} + \sum_{j=1}^{k-1} a_{i,k-j} \cdot \left(X^{\frac{d}{2k} \cdot (k-j)+i} - X^{\frac{d}{2k} \cdot (k+j)+i} \right) \\ + \sum_{j=1}^{k-1} a_{d/2k+i,k-j} \cdot \left(X^{\frac{d}{2k} \cdot (2k-j)+i} - X^{\frac{d}{2k} \cdot (2k+j)+i} \right). \end{aligned} \quad (9)$$

Then, the u -th coefficient of the polynomial above, where $0 \leq u < d/(2k)$, is exactly $a_{u,0}$. Similarly, the $(d/2k + u)$ -th coefficient is $a_{d/2k+u,0}$. Hence, $a_{u,0} = a_{d/2k+u,0} = 0$.

Now, we claim that for any $1 \leq u < k$ and $0 \leq v < \frac{d}{2k}$, $a_{v,u} = a_{d/2k+v,u} = 0$. Indeed, first consider the $(\frac{d}{2k} \cdot u + v)$ -th coefficient of the polynomial above. It is equal to $a_{v,u} + a_{d/2k+v,k-u} = 0$. On the other hand, if we focus on the $(\frac{d}{2k} \cdot (k+u) + v)$ -th coefficient, then it is exactly equal to $-a_{v,k-u} + a_{d/2k+v,u} = 0$. By combining the two observations we obtain

$$a_{d/2k+v,u} = a_{v,k-u} = -a_{d/2k+v,u}$$

which implies $a_{d/2k+v,u} = a_{v,u} = 0$, and thus $\mathbf{a} = \mathbf{0}$.

We move on to the second part. We will make use of the following three claims.

Claim. Consider the multiplicative group $\mathbb{Z}_{2d}^\times$ and let $k \leq d/2$ be a power-of-two. Then,

$$\langle 4k+1 \rangle = \{(4k) \cdot \alpha + 1 : 0 \leq \alpha \leq d/(2k) - 1\}.$$

Proof. First, using the property that $4k$ divides $2d$, we have for any i ,

$$((4k+1)^i \bmod 2d) \bmod 4k = (4k+1)^i \bmod 4k = \sum_{j=0}^i \binom{i}{j} (4k)^j \bmod 4k = 1.$$

Therefore, $\langle 4k+1 \rangle \subseteq \{(4k) \cdot \alpha + 1 : 0 \leq \alpha \leq d/(2k) - 1\}$. Now, the statement holds since by [LS18, Lemma 2.4], we know that the order of $4k+1 \in \mathbb{Z}_{2d}^\times$ is exactly $\frac{2d}{4k} = d/(2k)$. $\square$

Claim. Let $i \in \mathbb{Z}$ be not divisible by $d/2k$. Then, $\text{Tr}_H(X^i) = 0$.

Proof. Indeed, by assumption $4ki$ cannot be divisible $2d$ and thus $X^{4ki} \neq 1$. Now, using the first claim, we get

$$\begin{aligned} \text{Tr}_H(X^i) &= \sum_{h \in \langle -1, 4k+1 \rangle} X^{h \cdot i} \\ &= \sum_{h \in \langle 4k+1 \rangle} X^{h \cdot i} + X^{-h \cdot i} \\ &= \sum_{\alpha=0}^{d/(2k)-1} X^{(4k \cdot \alpha + 1) \cdot i} + \sum_{\alpha=0}^{d/(2k)-1} X^{-(4k \cdot \alpha + 1) \cdot i} \\ &= X^i \cdot \frac{(X^{4ki})^{d/(2k)} - 1}{X^{4ki} - 1} + X^{-i} \cdot \frac{(X^{-4ki})^{d/(2k)} - 1}{X^{-4ki} - 1} \\ &= 0. \end{aligned}$$

$\square$

Claim. $\text{Tr}_H(X^{d/2}) = 0$.

Proof. Using the first claim we obtain

$$\begin{aligned} \text{Tr}_H(X^{d/2}) &= \sum_{\alpha=0}^{d/(2k)-1} X^{(4k \cdot \alpha + 1) \cdot d/2} + \sum_{\alpha=0}^{d/(2k)-1} X^{-(4k \cdot \alpha + 1) \cdot d/2} \\ &= \sum_{\alpha=0}^{d/(2k)-1} X^{d/2} + \sum_{\alpha=0}^{d/(2k)-1} X^{-d/2} \\ &= \sum_{\alpha=0}^{d/(2k)-1} X^{d/2} - X^{d/2} = 0. \end{aligned}$$

$\square$

To conclude the proof of the theorem, we notice that

$$\begin{aligned} \psi(\mathbf{a}) \cdot \sigma_{-1}(\psi(\mathbf{b})) &= \left(\sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_i b_j \cdot X^{i-j} + \sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_{d/2k+i} b_{d/2k+j} \cdot X^{i-j} \right) \\ &\quad + X^{d/2} \cdot \left(\sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_{d/2k+i} b_j \cdot X^{i-j} - \sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_i b_{d/2k+j} \cdot X^{i-j} \right). \end{aligned}$$

Hence,

$$\begin{aligned} \text{Tr}_H(\psi(\mathbf{a}) \cdot \sigma_{-1}(\psi(\mathbf{b}))) &= \left(\sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_i b_j \cdot \text{Tr}_H(X^{i-j}) + \sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_{d/2k+i} b_{d/2k+j} \cdot \text{Tr}_H(X^{i-j}) \right) \\ &\quad + \left(\sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_{d/2k+i} b_j \cdot \text{Tr}_H(X^{d/2+(i-j)}) - \sum_{i=0}^{d/2k-1} \sum_{j=0}^{d/2k-1} a_i b_{d/2k+j} \cdot \text{Tr}_H(X^{d/2+(i-j)}) \right). \end{aligned}$$

Now, using the fact that for two distinct $0 \leq i, j < d/2k$, both $i - j$ and $d/2 + (i - j)$ cannot be divisible by $d/2k$, and the second claim, we deduce that $\text{Tr}_H(X^{i-j}) = \text{Tr}_H(X^{d/2+(i-j)}) = 0$. However, when $i = j$, then $\text{Tr}_H(X^{i-j}) = \frac{d}{k}$ and $\text{Tr}_H(X^{d/2+(i-j)}) = 0$ by the third claim. Hence, we deduce that

$$\text{Tr}_H(\psi(\mathbf{a}) \cdot \sigma_{-1}(\psi(\mathbf{b}))) = \frac{d}{k} \sum_{i=0}^{d/2k-1} a_i b_i = \frac{d}{k} \langle \mathbf{a}, \mathbf{b} \rangle$$

which concludes the proof. $\square$

Next, we give a simple observation regarding how infinity norm changes after applying the ψ map.

Lemma 6. *Let $\mathbf{a}$ be a vector over $\mathbf{R}_q^H$ such that $\|\mathbf{a}\|_\infty \leq \beta$ and ψ be the map defined in Theorem 2. Then, $\|\psi(\mathbf{a})\|_\infty \leq 2\beta$.*

Proof. The statement follows directly from the explicit formula in Equation (9), where some coefficients of $\psi(\mathbf{a})$ are equal to the sum of two coefficients of $\mathbf{a}$.

Reducing to multilinear evaluation over $\mathbf{R}_q$. We use the results above to translate the multilinear evaluation claim over $\mathbf{R}_q^H$

$$f(x_1, \dots, x_\ell) = y$$

to an equivalent relation over the ring $\mathbf{R}_q$. Denote the coefficients of polynomial f as $(f_i)_{i \in \{0,1\}^\ell} \in \mathbf{R}_q^H$. We can explicitly rewrite the evaluation equation as

$$y = \sum_{i \in \{0,1\}^{\ell-\alpha+\kappa}} \left(x_1^{i_1} \cdot \dots \cdot x_{\ell-\alpha+\kappa}^{i_{\ell-\alpha+\kappa}} \right) \cdot \left(\sum_{j \in \{0,1\}^{\alpha-\kappa}} f_{i||j} \cdot x_{\ell-\alpha+\kappa+1}^{j_1} \cdot \dots \cdot x_\ell^{j_{\alpha-\kappa}} \right). \quad (10)$$

For each $i \in \{0,1\}^{\ell-\alpha+\kappa}$ we define

$$F_i := \psi \left((f_{i||j})_{j \in \{0,1\}^{\alpha-\kappa}} \right) \quad \text{and also} \quad v := \psi \left((x_{\ell-\alpha+\kappa+1}^{j_1} \cdot \dots \cdot x_\ell^{j_{\alpha-\kappa}})_{j \in \{0,1\}^{\alpha-\kappa}} \right).$$

Now, recall that for every $x \in \mathbf{R}_q^H$ we have $\text{Tr}_H(x) = x$ and trace is additively homomorphic. Hence, by applying Theorem 2 to Equation (10) we obtain

$$\frac{d}{k} \cdot y = \text{Tr}_H \left(\underbrace{\sum_{i \in \{0,1\}^{\ell-\alpha+\kappa}} \left(x_1^{i_1} \cdot \dots \cdot x_{\ell-\alpha+\kappa}^{i_{\ell-\alpha+\kappa}} \right) \cdot F_i}_{Y:=} \cdot \sigma_{-1}(v) \right).$$

Therefore, the prover will send a single ring element $Y \in \mathbf{R}_q$, which allows the verifier to check if $\text{Tr}_H(Y \cdot \sigma_{-1}(v)) = dy/k$. Then, we only need to prove that Y is well-formed, which is an evaluation proof for the $(\ell - \alpha + \kappa)$ -variate multilinear polynomial $F := (F_i)_i$ over $\mathbf{R}_q$.

3.2 Dealing with $f \in \mathbb{Z}_q^{\leq 1}[X_1, \dots, X_\ell]$

We now optimize the special case where the evaluation point $(x_1, \dots, x_\ell)$ is defined in the extension field $\mathbb{F}_{q^k}$, while the coefficients of the committed polynomial f are over $\mathbb{Z}_q$. Note that if we naively apply the generic transformation from Section 3.1 then we would end up with an evaluation proof for the $(\ell - \alpha + \kappa)$ -variate multilinear polynomial over $\mathbf{R}_q$. This subsection focuses on how to lower it further to $\ell - \alpha$ variables.

We start with a similar observation as in Equation (10) that a polynomial evaluation claim can be written as

$$y = \sum_{i \in \{0,1\}^\kappa} \left(x_1^{i_1} \cdot \dots \cdot x_\kappa^{i_\kappa} \right) \cdot \underbrace{\left(\sum_{j \in \{0,1\}^{\ell-\kappa}} f_{i||j} \cdot x_{\kappa+1}^{j_1} \cdot \dots \cdot x_\ell^{j_{\ell-\kappa}} \right)}_{y_i :=}. \quad (11)$$

Hence, we let the prover send k partial evaluations $y_i \in \mathbb{F}_{q^k}$ to the verifier who can directly check whether

$$y = \sum_i \left(x_1^{i_1} \cdot \dots \cdot x_\kappa^{i_\kappa} \right) \cdot y_i.$$

What we have left to prove is that all the y_i are well-formed, or in other words $f_i(x_{\kappa+1}, \dots, x_\ell) = y_i$ for $i \in \{0,1\}^\kappa$ where $f_i \in \mathbb{Z}_q[X_{\ell-\kappa+1}, \dots, X_\ell]$.

For the next part, we need a somewhat explicit definition of the extension field (it could as well be $\mathbf{R}_q^H$). Let $\mathbb{F}_{q^k} := \mathbb{F}_q[Z]/\varphi(Z)$, where φ is an irreducible polynomial in $\mathbb{F}_q[Z]$ of degree k . Then, we define the $(\ell - \kappa)$ -variate multilinear polynomial $f' \in \mathbb{F}_{q^k}[X_{\kappa+1}, \dots, X_\ell]$ as

$$f'(X_{\kappa+1}, \dots, X_\ell) := \sum_{i \in \{0,1\}^\kappa} f_i(X_{\kappa+1}, \dots, X_\ell) \cdot Z^{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}}.$$

Then, by definition of the partial evaluations y_i we obtain

$$f'(x_{\kappa+1}, \dots, x_\ell) = \sum_{i \in \{0,1\}^\kappa} y_i \cdot Z^{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \in \mathbb{F}_{q^k}$$

which is now a full-fledged polynomial evaluation over $\mathbb{F}_{q^k}$. By applying the generic transformation from Section 3.1, we can reduce the statement into an evaluation claim over $\mathbf{R}_q$ for a μ -variate multilinear polynomial, where

$$\mu = \underbrace{(\ell - \kappa)}_{\text{Section 3.2}} - \alpha + \kappa = \ell - \alpha.$$

Section 3.1

The final observation is that the verifier can directly compute $y_{0\dots 0}$ from y and other y_i , and thus only $k - 1$ evaluations are needed to be sent.

Remark 1. One may notice a strange behavior in Figure 2 in a sense that the transformation from a field extension $\mathbb{F}_{q^k}$ to $\mathbf{R}_q$ seems cheaper than from the base field $\mathbb{F}_q$ to $\mathbf{R}_q$. Indeed, if given a witness polynomial f over $\mathbb{F}_q$, why is it not enough to just follow the transformation from Section 3.1?

There are two main reasons. The first one is that, when combined with a polynomial evaluation protocol over $\mathbf{R}_q$, it is not guaranteed in the knowledge soundness proof that the extracted polynomial f is also over $\mathbb{F}_q$, as we obtain $f \in \mathbb{F}_{q^k}[X_1, \dots, X_\ell]$. Moreover, the transformation in Section 3.2 allows us to further reduce the number of variables of the $\mathbf{R}_q$ -polynomial, from $\ell - \alpha + \kappa$ to $\ell - \alpha$, which has significant impact in reducing round complexity of the protocol.

Committed polynomial	# variables after the $\mathbf{R}_q$ -transformation	Prover cost
$f \in \mathbb{F}_{q^k}[X_1, \dots, X_\ell]$	$\ell - \alpha + \kappa$	$d \log q$
$f \in \mathbb{F}_q[X_1, \dots, X_\ell]$	$\ell - \alpha$	$(d + k(k-1)) \log q$

Fig. 2. Summary of our transformation in Sections 3.1 and 3.2. The prover cost is measured in the size of the prover messages in bits.

4 Multilinear Polynomial Commitment Scheme over $\mathbf{R}_q$

This section focuses on proving knowledge of a *committed* multilinear polynomial $f \in \mathbf{R}_q^{\leq 1}[X_1, \dots, X_\ell]$ with coefficients $(f_i)_{i \in \{0,1\}^\ell}$ such that

$$f(x_1, \dots, x_\ell) := \sum_{i \in \{0,1\}^\ell} f_i \cdot x_1^{i_1} \cdot \dots \cdot x_\ell^{i_\ell} = u$$

where the evaluation point $(x_1, \dots, x_\ell) \in \mathbf{R}_q^\ell$ and the image $u \in \mathbf{R}_q$ are public.

We follow the initial strategy from Greyhound [NS24]. Suppose $\ell = m + r$ for some $m, r \geq 1$ (keep in mind $m \approx r$). Then, we can rewrite the equation above as

$$f(x_1, \dots, x_\mu) := \sum_{i \in \{0,1\}^r} \sum_{j \in \{0,1\}^m} f_{i||j} \cdot (x_1^{i_1} \cdot \dots \cdot x_r^{i_r}) \cdot (x_{r+1}^{j_1} \cdot \dots \cdot x_\ell^{j_m}),$$

or in other words

$$f(x_1, \dots, x_\mu) = \mathbf{b}^\top \begin{bmatrix} \mathbf{a}^\top & & \\ & \ddots & \\ & & \mathbf{a}^\top \end{bmatrix} \begin{bmatrix} \mathbf{f}_{0\dots 0} \\ \vdots \\ \mathbf{f}_{1\dots 1} \end{bmatrix} \quad \text{for } \begin{array}{l} \mathbf{b}^\top := (x_1^{i_1} \cdot \dots \cdot x_r^{i_r})_{i \in \{0,1\}^r} \in \mathbf{R}_q^{2^r} \\ \mathbf{a}^\top := (x_{r+1}^{j_1} \cdot \dots \cdot x_\ell^{j_m})_{j \in \{0,1\}^m} \in \mathbf{R}_q^{2^m} \\ \mathbf{f}_i^\top := (f_{i||j})_{j \in \{0,1\}^m} \in \mathbf{R}_q^{2^m} \text{ for } i \in \{0,1\}^r \end{array}. \quad (12)$$

This representation of a polynomial evaluation claim gives intuition how to commit to the polynomial f .

4.1 Inner and Outer Commitment

The starting point of our polynomial commitment scheme construction is the basic commitment scheme from LaBRADOR [BS23]. Let $b > 1$ be the base used for decomposition (throughout) and denote $\delta := \lceil \log_b q \rceil$. The commitment key is a pair of uniformly random matrices $\mathbf{A} \in \mathbf{R}_q^{n_A \times \delta 2^m}$ and $\mathbf{B} \in \mathbf{R}_q^{n_B \times n_A \delta 2^r}$ where n_A, n_B are the security parameters which determine hardness of the underlying Module-SIS problem.

Suppose we want to commit to the 2^r vectors $(\mathbf{f}_i^\top)_i$ of length 2^m . The first step is to compute the decompositions:

$$\mathbf{s}_{1+\sum_{z=1}^r i_z \cdot 2^{z-1}} := \mathbf{G}_{2^m}^{-1}(\mathbf{f}_i) \quad (13)$$

and the inner Ajtai commitments $\mathbf{t}_i := \mathbf{A} \mathbf{s}_i \in \mathbf{R}_q^{n_A}$ for $i \in [2^r]$. The next step is to compute their binary decomposition $\hat{\mathbf{t}}_i := \mathbf{G}_{n_A}^{-1}(\mathbf{t}_i)$. The final outer commitment is

$$\mathbf{u} := \mathbf{B} \begin{bmatrix} \hat{\mathbf{t}}_1 \\ \vdots \\ \hat{\mathbf{t}}_{2^r} \end{bmatrix} \in \mathbf{R}_q^{n_B}. \quad (14)$$

The commitment opening becomes a tuple of $(\mathbf{s}_i, \hat{\mathbf{t}}_i)_i$.

In the knowledge extraction, we will only be able to extract so-called weak openings. We define a weak opening for the commitment $\mathbf{u}$ is a tuple $(\mathbf{s}_i, \hat{\mathbf{t}}_i, c_i)_{i \in [2^r]}$, which satisfies all the following conditions

$$\begin{aligned} \forall i \in [2^r] : \quad & \|c_i \cdot \mathbf{s}_i\| \leq \bar{\beta}, \quad \|c_i\|_1 \leq \bar{\omega}, \quad c_i \in \mathbf{R}_q^\times, \quad \mathbf{A}\mathbf{s}_i = \mathbf{G}_{n_B} \hat{\mathbf{t}}_i \\ \mathbf{B} \begin{bmatrix} \hat{\mathbf{t}}_1 \\ \vdots \\ \hat{\mathbf{t}}_{2^r} \end{bmatrix} = \mathbf{u} \quad & \text{and} \quad \left\| \begin{bmatrix} \hat{\mathbf{t}}_1 \\ \vdots \\ \hat{\mathbf{t}}_{2^r} \end{bmatrix} \right\|_\infty \leq \bar{\gamma}. \end{aligned}$$

Clearly, a message can be derived from the weak opening by Equation (13). We recall the known result that the commitment scheme above satisfies binding w.r.t. weak openings under the Module-SIS assumption. We omit the proof since it follows identically as in [NS24, Lemma 2.11].

Lemma 7 (Weak Binding). *There is a deterministic algorithm that given two weak openings $(\mathbf{s}_i, \hat{\mathbf{t}}_i, c_i)_{i \in [2^r]}$ and $(\mathbf{s}'_i, \hat{\mathbf{t}}'_i, c'_i)_{i \in [2^r]}$ for the commitment $\mathbf{u} \in \mathbf{R}_q^{n_B}$ such that $\mathbf{s}_j \neq \mathbf{s}'_j$ for some $j \in [2^r]$, outputs a vector $\mathbf{z} \in \mathbf{R}_q^{2^m + n_A \delta 2^r}$ such that $[\mathbf{A} \mid \mathbf{B}]\mathbf{z} = \mathbf{0}$ and $0 < \|\mathbf{z}\|_\infty \leq \max(2\bar{\omega}\bar{\beta}, 2\bar{\gamma})$.*

4.2 Polynomial Evaluation as Quadratic Equation

We can now rephrase Equation (12) in terms of the commitment opening $(\mathbf{s}_i)_i$. Namely, we can write

$$u = f(x_1, \dots, x_\ell) = \mathbf{b}^\top \begin{bmatrix} \mathbf{a}^\top \mathbf{G}_{2^m} & & \\ & \ddots & \\ & & \mathbf{a}^\top \mathbf{G}_{2^m} \end{bmatrix} \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_{2^r} \end{bmatrix}. \quad (15)$$

From now on, we focus on proving knowledge of a commitment opening $(\mathbf{s}_i, \hat{\mathbf{t}}_i)_i$ for the commitment $\vec{\mathbf{u}}$ which satisfies the above equation. We summarize the protocol in Figure 3.

To begin with, the prover commits to the vector $\mathbf{w} = (w_1, \dots, w_{2^r}) \in \mathbf{R}_q^{2^r}$ defined as $w_i := \mathbf{a}^\top \mathbf{G}_{2^m} \mathbf{s}_i \in \mathbf{R}_q$. That is, $\mathcal{P}$ first decomposes each w_i , thus obtaining $\hat{\mathbf{w}}_i := \mathbf{G}_1^{-1}(w_i) \in \mathbf{R}_q^\delta$. Finally, the prover sends the commitment

$$\mathbf{v} := \mathbf{D}\hat{\mathbf{w}} \in \mathbf{R}_q^{n_D} \quad (16)$$

where $\mathbf{D} \in \mathbf{R}_q^{n_D \times \delta \cdot 2^r}$ is a public uniformly random matrix and $\hat{\mathbf{w}}^\top := (\hat{\mathbf{w}}_1^\top, \dots, \hat{\mathbf{w}}_{2^r}^\top)$. Note that Equation (15) can be rephrased as

$$u = \mathbf{b}^\top \mathbf{w} = \mathbf{b}^\top \mathbf{G}_{2^r} \hat{\mathbf{w}}. \quad (17)$$

The verifier $\mathcal{V}$ responds with a challenge vector $\mathbf{c} := (c_1, \dots, c_{2^r}) \leftarrow \mathcal{C}^{2^r}$, where the challenge space $\mathcal{C}$ is currently defined as a subset of short challenges in $\mathbf{R}_q$:

$$\mathcal{C} \subseteq \{c \in \mathbf{R}_q : \|c\|_1 \leq \omega\}$$

for some ω instantiated later⁹. Then, the prover $\mathcal{P}$ computes the short folded witness

$$\mathbf{z} := c_1 \mathbf{s}_1 + c_2 \mathbf{s}_2 + \dots + c_{2^r} \mathbf{s}_{2^r} \in \mathbf{R}_q^{2^m \delta}.$$

We note the relationship between $\mathbf{z}$ and $\hat{\mathbf{w}}$

$$\mathbf{a}^\top \mathbf{G}_{2^m} \mathbf{z} = \sum_{i=1}^{2^r} c_i \mathbf{a}^\top \mathbf{G}_{2^m} \mathbf{s}_i = \sum_{i=1}^{2^r} c_i w_i = \sum_{i=1}^{2^r} c_i \mathbf{G}_1 \hat{\mathbf{w}}_i = (\mathbf{c}^\top \otimes \mathbf{G}_1) \hat{\mathbf{w}}. \quad (18)$$

We also note the relationship between $\mathbf{z}$ and $\hat{\mathbf{t}}$:

$$\mathbf{A}\mathbf{z} := \sum_{i=1}^{2^r} c_i \mathbf{A}\mathbf{s}_i = \sum_{i=1}^{2^r} c_i \mathbf{G}_{n_A} \hat{\mathbf{t}}_i = (\mathbf{c} \otimes \mathbf{G}_{n_A}) \hat{\mathbf{t}}. \quad (19)$$

⁹ The ℓ_1 norm bound for challenges is set here to later apply Lemma 2.

Now, combining Equations 14, 16, 17, 18, and 19, the prover $\mathcal{P}$ will prove that short $\mathbf{z}, \hat{\mathbf{t}}, \hat{\mathbf{w}}$ satisfy the linear relation over $\mathbf{R}_q$:

$$\begin{bmatrix} \mathbf{D} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} & \mathbf{0} \\ \mathbf{b}^\top \mathbf{G}_{2^r} & \mathbf{0} & \mathbf{0} \\ (\mathbf{c}^\top \otimes \mathbf{G}_1) & \mathbf{0} & -\mathbf{a}^\top \mathbf{G}_{2^m} \\ \mathbf{0} & \mathbf{c}^\top \otimes \mathbf{G}_{n_A} & -\mathbf{A} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{w}} \\ \hat{\mathbf{t}} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{u} \\ u \\ 0 \\ 0 \end{bmatrix}.$$

We see that while coefficients of $\hat{\mathbf{w}}, \hat{\mathbf{t}}$ are between $[-b/2]$ and $[b/2] - 1$, the coefficients of $\mathbf{z}$ will be moderately larger. Let β be the maximum L_∞ norm of $\mathbf{z}$, $\tau := \lceil \log_b \beta \rceil$ and $\mathbf{J}_n := \mathbf{I}_n \otimes [1 \ b \ b^2 \ \dots \ b^{\tau-1}] \in \mathbf{R}_q^{n \times n\tau}$ be the corresponding gadget matrix. To decrease the coefficients of $\mathbf{z}$ further, we decompose as $\hat{\mathbf{z}} = \mathbf{J}_{2^m}^{-1}(\mathbf{z}) \in \mathbf{R}_q^{2^m \delta \tau}$. Hence, the prover now needs to prove knowledge of $(\hat{\mathbf{w}}, \hat{\mathbf{t}}, \hat{\mathbf{z}}) \in S_b^{2^r \delta + 2^r n_A \delta + 2^m \delta \tau}$ such that

$$\begin{bmatrix} \mathbf{D} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} & \mathbf{0} \\ \mathbf{b}^\top \mathbf{G}_{2^r} & \mathbf{0} & \mathbf{0} \\ (\mathbf{c}^\top \otimes \mathbf{G}_1) & \mathbf{0} & -\mathbf{a}^\top \mathbf{G}_{2^m} \mathbf{J}_{2^m} \\ \mathbf{0} & \mathbf{c}^\top \otimes \mathbf{G}_{n_A} & -\mathbf{A} \mathbf{J}_{2^m} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{w}} \\ \hat{\mathbf{t}} \\ \hat{\mathbf{z}} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{u} \\ u \\ 0 \\ 0 \end{bmatrix}. \quad (20)$$

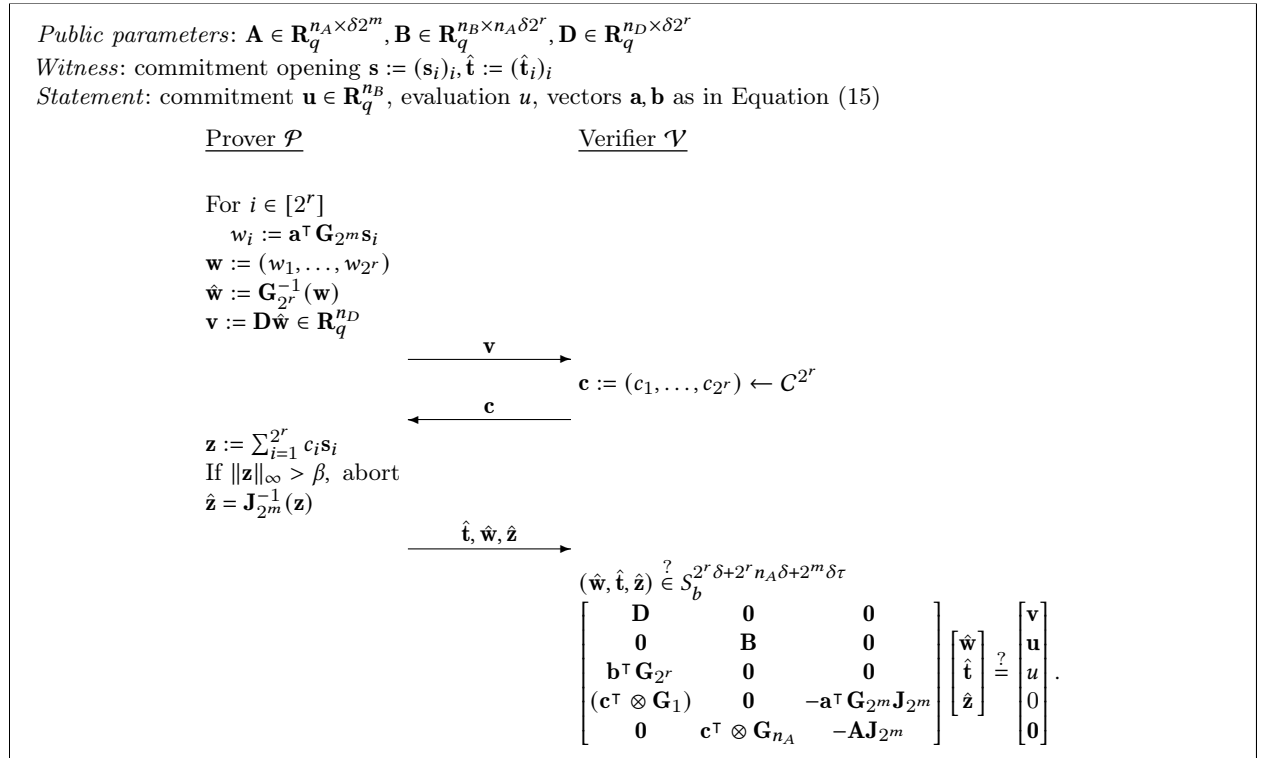


Fig. 3. Interactive proof of knowledge of the commitment opening $(\mathbf{s}_i, \hat{t}_i)$ which satisfies (15). In our final scheme, the prover does not send the final message in the clear, but proves knowledge of valid $(\hat{\mathbf{w}}, \hat{\mathbf{t}}, \hat{\mathbf{z}})$.

The completeness follows directly from the protocol rationale described above. Hence, we move on the soundness guarantees. Identically as [NS24, Lemma 3.2] we show that given CWSS-structured transcripts, we can extract a valid weak opening which satisfies Equation (15).

Lemma 8 (CWSS of Figure 3). Let $(\bar{\beta}, \bar{\omega}, \bar{\gamma}) := (2b^k, 2\omega, b)$. Suppose $\omega < \frac{1}{2\sqrt{2}}q^{1/2}$. Then, given the public parameters, the statement $(\mathbf{u}, \mathbf{u})$, and $2^r + 1$ valid transcripts of the form

$$\text{tr}^{(j)} := (\mathbf{v}, \mathbf{c}^{(j)}, (\hat{\mathbf{w}}^{(j)}, \hat{\mathbf{t}}^{(j)}, \hat{\mathbf{z}}^{(j)})) \quad \text{where} \quad (\mathbf{c}^{(j)})_{0 \leq j \leq 2^r} \in \text{SS}(C, 2^r, 2),$$

one can either efficiently extract a weak opening $(\bar{\mathbf{s}}^{(j)}, \bar{\mathbf{t}}^{(j)}, \mathbf{c}^{(j)})_{j \in [2^r]}$ for the commitment $\mathbf{u}$ such that

$$\mathbf{u} = \mathbf{b}^\top \begin{bmatrix} \mathbf{a}^\top \mathbf{G}_{2^m} & & \\ & \ddots & \\ & & \mathbf{a}^\top \mathbf{G}_{2^m} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{s}}_1 \\ \vdots \\ \bar{\mathbf{s}}_{2^r} \end{bmatrix},$$

or solve $\text{MSIS}_{q,d,n_B,\delta 2^r n_A, 2b}$ or $\text{MSIS}_{q,d,n_D,\delta 2^r, 2b}$.

Proof. For each $j \in [2^r]$, let us reconstruct the vector $\mathbf{z}^{(j)} := \mathbf{J}_{2^m} \hat{\mathbf{z}}^{(j)}$. Then, we have $\|\mathbf{z}^{(j)}\|_\infty \leq b^k$.

Without loss of generality, assume that $\mathbf{c}^{(0)}$ is the “central vector”. First, suppose that for two distinct $i \neq j$ we have $\hat{\mathbf{t}}^{(i)} \neq \hat{\mathbf{t}}^{(j)}$. In particular, we have $\mathbf{B}\hat{\mathbf{t}}^{(i)} = \mathbf{u} = \mathbf{B}\hat{\mathbf{t}}^{(j)}$, and so we have found a short solution of norm at most $2b$ for the matrix $\mathbf{B}$. Similarly, we argue for the vectors $\hat{\mathbf{w}}^{(j)}$. Hence, from now on we assume that all $\hat{\mathbf{t}}^{(j)} := \hat{\mathbf{t}}$ and $\hat{\mathbf{w}}^{(j)} := \hat{\mathbf{w}}$ are the same. Denote $\mathbf{w} := \mathbf{G}_{2^r} \hat{\mathbf{w}}$ and parse $\mathbf{w} = (\mathbf{w}_i)_i$, $\hat{\mathbf{t}} := (\hat{\mathbf{t}}_i)_i$. Then, we have $\mathbf{b}^\top \mathbf{w} = \mathbf{u}$.

Fix $j \in [2^r]$ and consider the transcripts $\text{tr}^{(j)}$ and $\text{tr}^{(0)}$ with the verifier challenges denoted as $\mathbf{c}^{(j)} := (c_{j,i})_i$ and $\mathbf{c}^{(0)} := (c_{0,i})_i$. Recall that the two challenge vectors differ *exactly* in the j -th coordinate and $c_{j,i} = c_{0,i}$ for all $i \neq j$. By subtracting the two corresponding verification equations, we obtain

$$\mathbf{b}^\top \mathbf{G}_{2^m} (\mathbf{z}^{(j)} - \mathbf{z}^{(0)}) = (c_{j,j} - c_{0,j}) \cdot \mathbf{w}_j \quad \text{and} \quad \mathbf{A}(\mathbf{z}^{(j)} - \mathbf{z}^{(0)}) = (c_{j,j} - c_{0,j}) \cdot \mathbf{G}_{b,n} \hat{\mathbf{t}}_j.$$

Hence, define our extracted opening as $\bar{\mathbf{s}}\mathbf{t} := (\mathbf{s}_j, \hat{\mathbf{t}}_j)_j$ where

$$\mathbf{s}_j := \frac{\mathbf{z}^{(j)} - \mathbf{z}^{(0)}}{c_{j,j} - c_{0,j}} \quad \text{for all } j \in \{0, 1\}^r$$

and the slack vector $(\bar{c}_j)_j$ is defined as $\bar{c}_j := c_{j,j} - c_{0,j}$. By construction, we have $\|\bar{c}_j\|_1 \leq 2\omega$ and $\|\bar{c}_j \mathbf{s}_j\|_\infty \leq 2b^k$. By Lemma 3, $\bar{c}_j$ is invertible over $\mathbf{R}_q$ since $\|\bar{c}_j\|_\infty \leq 2\omega < \frac{1}{\sqrt{2}}q^{1/2}$. Finally, $\mathbf{b}^\top \mathbf{G}_{2^m} \mathbf{s}_j = \mathbf{w}_j$ for all $j \in [2^r]$, which concludes the proof. $\square$

4.3 Unstructured Linear Relations and Sumcheck

In this section, we show how to reduce proving unstructured linear relations over $\mathbf{R}_q$ with infinity-norm constraints, as in Equation (20). Concretely, we focus of the following relation $\mathbf{R}^{\text{lin}}$:

$$\mathbf{R}_{q,d,n,\mu,b}^{\text{lin}} := \{(\mathbf{z} \in \mathbf{R}_q^\mu, (\mathbf{M} \in \mathbf{R}_q^{n \times \mu}, \mathbf{y} \in \mathbf{R}_q^n)) : \mathbf{M}\mathbf{z} = \mathbf{y} \wedge \|\mathbf{z}\|_\infty \leq b - 1\}.$$

At a high level, our approach is to commit to $\mathbf{z}$ using a (potentially another) polynomial commitment scheme, and then apply the sumcheck protocol to prove the relation. However, one challenge is that the relation $\mathbf{R}_{q,d,n,\mu,b}^{\text{lin}}$ is defined over polynomial rings $\mathbf{R}_q$, so the standard sumcheck protocol over finite fields cannot be directly applied (or one would have to run it over $\mathbf{R}_q$ as in [BC24]). To bridge this gap, we leverage the ring-switching technique introduced by [HMZ25].

Ring switching. In the relation $\mathbf{R}_{q,d,n,\mu,b}^{\text{lin}}$, the input $\mathbf{M} \in \mathbf{R}_q^{n \times \mu}$ and $\mathbf{y} \in \mathbf{R}_q^n$ are public information, and the prover needs to prove knowledge of a witness $\mathbf{z} \in \mathbf{R}_q^\mu$ such that

- $\mathbf{M}\mathbf{z} = \mathbf{y}$
- $\|\mathbf{z}\|_\infty \leq b - 1$

For the first identity, the authors of [HMZ25] observe that $\mathbf{M}\mathbf{z} = \mathbf{y}$ if and only if there is a polynomial $\mathbf{r} \in (\mathbb{Z}_q^{<d}[X])^n$ such that

$$\mathbf{M}\mathbf{z} = \mathbf{y} + (X^d + 1) \cdot \mathbf{r}$$

We now naturally consider $\mathbf{M}$, $\mathbf{z}$, and $\mathbf{y}$ as matrices or vectors over $\mathbb{Z}_q^{<d}[X]$ (instead of over $\mathbf{R}_q$). Since we will apply the sumcheck protocol later, we further consider $\mathbf{M}, \mathbf{x}, \mathbf{y}, \mathbf{r}$ as elements over $\mathbb{F}_{q^k}^{<d}[X]$. Now, we can divide the process into two steps: the prover first commits to $(\mathbf{z}, \mathbf{r})$ using a polynomial commitment scheme Com , and then proves the following constraints:

- For each $i \in [n]$, $\sum_j \mathbf{M}_{i,j} \mathbf{z}_j = \mathbf{y}_i + (X^d + 1) \cdot \mathbf{r}_i$, where $\mathbf{M}_{i,j}, \mathbf{y}_i \in \mathbb{Z}_q^{<d}[X]$
- $\mathbf{z}$ is a vector over $\mathbb{Z}_q^{<d}[X]$ and $\|\mathbf{z}\|_\infty \leq b - 1$.

Notice that the verifier can check whether $\mathbf{M}$ and $\mathbf{y}$ are over $\mathbb{Z}_q^{<d}[X]$ by themselves, since $\mathbf{M}$ and $\mathbf{y}$ are public information. However, the prover has to prove that $\mathbf{z}$ is a vector over $\mathbb{Z}_q^{<d}[X]$ if we run the sumcheck protocol over $\mathbb{F}_{q^k}^{<d}[X]$. Since $\mathbf{M}_{i,j}, \mathbf{z}_j, \mathbf{y}_i, \mathbf{r}_i \in \mathbb{Z}_q^{<d}[X]$, we can represent them as

$$\mathbf{M}_{i,j} = \sum_{\ell < d} M_{i,j,\ell} \cdot X^\ell,$$

and similarly for all $\mathbf{z}_j, \mathbf{y}_i$, and $\mathbf{r}_i$. Here we note that both $\mathbf{z}$ and $\mathbf{r}$ are witnesses of the prover, and the prover needs to prove the norm bound of $\mathbf{z}$. For the sake of notation and implementation simplicity, we also let the prover performs a gadget decomposition of $\mathbf{r}$ as $\mathbf{r} = \sum_u b^u \cdot \mathbf{r}_u$, and commits $(\mathbf{z}, \mathbf{r}_1, \dots, \mathbf{r}_{\log_b(q)})$ instead, and also proves the norm bound of $\mathbf{r}_u$ for each u . That is, the prover proves the following statement:

- For each $i \in [n]$, $\sum_j \mathbf{M}_{i,j} \mathbf{z}_j = \mathbf{y}_i + (X^d + 1) \cdot \sum_u b^u \mathbf{r}_{i,u}$, where $\mathbf{M}_{i,j}, \mathbf{y}_i \in \mathbb{Z}_q^{<d}[X]$
- $\mathbf{z}, \mathbf{r}_1, \dots, \mathbf{r}_{\log_b(q)}$ are vectors over $\mathbb{Z}_q^{<d}[X]$ and $\|\mathbf{z}\|_\infty, \|\mathbf{r}_1\|_\infty, \dots, \|\mathbf{r}_{\log_b(q)}\|_\infty \leq b - 1$.

To simplify the presentation, in the following we omit the subscript u and explain how to prove the statement below. Readers should be aware that there is a hidden gadget decomposition of $\mathbf{r}$. We believe this is sufficient to explain our main idea. To incorporate this gadget decomposition, we can modify the multilinear extension correspondingly.

- For each $i \in [n]$, $\sum_j \mathbf{M}_{i,j} \mathbf{z}_j = \mathbf{y}_i + (X^d + 1) \cdot \mathbf{r}_i$, where $\mathbf{M}_{i,j}, \mathbf{y}_i \in \mathbb{Z}_q^{<d}[X]$
- $\mathbf{z}, \mathbf{r}$ are vectors over $\mathbb{Z}_q^{<d}[X]$ and $\|\mathbf{z}\|_\infty, \|\mathbf{r}\|_\infty \leq b - 1$.

For each $\alpha \in \mathbb{F}_{q^k}$, we denote

$$\mathbf{M}_{i,j}(\alpha) = \sum_{\ell < d} M_{i,j,\ell} \cdot \alpha^\ell \in \mathbb{F}_{q^k},$$

and similarly for all $\mathbf{z}_j, \mathbf{y}_i$, and $\mathbf{r}_i$. Now, instead of proving the constraints over $\mathbb{F}_{q^k}^{<d}[X]$, which could be very costly, we optimize by letting the verifier first send a random α to the prover. Then, the prover proves the following constraints:

- For each $i \in [n]$, $\sum_j \mathbf{M}_{i,j}(\alpha) \mathbf{z}_j(\alpha) = \mathbf{y}_i(\alpha) + (\alpha^d + 1) \cdot \mathbf{r}_i(\alpha)$, where $\mathbf{M}_{i,j}, \mathbf{y}_i \in \mathbb{Z}_q^{<d}[X]$
- $\mathbf{z}$ is a vector over $\mathbb{Z}_q^{<d}[X]$ and $\|\mathbf{z}\|_\infty, \|\mathbf{r}\|_\infty \leq b - 1$.

Having proof composition in mind, we consider the following simple three-round protocol in Figure 4 which captures the idea above.

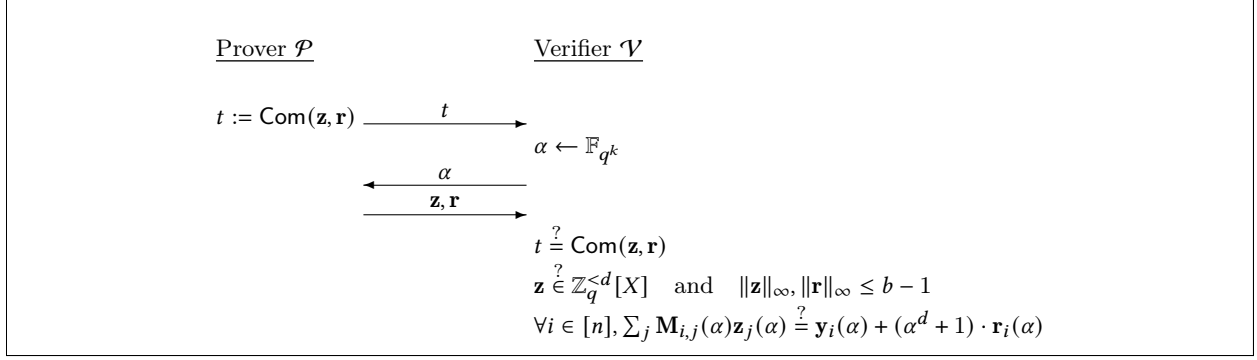


Fig. 4. Reducing proving linear equations over $\mathbb{Z}_q^{<d}[X]$ to equations over $\mathbb{F}_{q^k}$.

We note that this simplification could introduce a $(2d-1)/|\mathbb{F}_{q^k}|$ soundness error. This is because the polynomial $\sum_j \mathbf{M}_{i,j} \mathbf{z}_j - \mathbf{y}_i - (X^d + 1) \cdot \mathbf{r}_i$ has degree at most $2d-1$. Formally, we argue this by proving $2d$ -special soundness in Lemma 9.

Lemma 9 (Special soundness of Figure 4). *There is an efficient, deterministic algorithm that given a statement $(\mathbf{M}_{i,j}, \mathbf{y}_i, b)$ and $2d$ valid transcripts $(t_i, \alpha_i, (\mathbf{z}_i, \mathbf{r}_i))$ of the protocol in Figure 4 such that $(\alpha_i)_i \in \text{SS}(\mathbb{F}_{q^k}, 1, 2d)$, can either extract $\bar{\mathbf{z}}, \bar{\mathbf{r}}$ which satisfy $\mathbf{M}\mathbf{z} = \mathbf{y} + (X^d + 1) \cdot \mathbf{r}$ over $\mathbb{Z}_q[X]$, or break binding of the commitment scheme Com.*

Proof. If for some $i \neq j$ we have $(\mathbf{z}_i, \mathbf{r}_i) \neq (\mathbf{z}_j, \mathbf{r}_j)$, then we automatically break the binding property of the commitment scheme. Otherwise, define $\bar{\mathbf{z}} := \mathbf{z}_i, \bar{\mathbf{r}} := \mathbf{r}_i$ for all i . Then, for $2d$ distinct evaluation points α_j , we have

$$\forall i \in [n], \sum_j \mathbf{M}_{i,j}(\alpha_j) \mathbf{z}_j(\alpha_j) \stackrel{?}{=} \mathbf{y}_i(\alpha_j) + (\alpha_j^d + 1) \cdot \mathbf{r}_i(\alpha_j).$$

Since the polynomial $\sum_j \mathbf{M}_{i,j} \mathbf{z}_j - \mathbf{y}_i - (X^d + 1) \cdot \mathbf{r}_i$ has degree at most $2d-1$, this implies that it must be a zero polynomial, which finishes the proof. $\square$

From now on, we focus on proving knowledge of the prover's last message which satisfies the verification conditions.

Remark 2. Looking ahead, we will instantiate Com with the inner-outer commitment from Section 4. In this case, for special soundness we will consider breaking the weak binding property (cf. Lemma 7), rather than binding.

Represent the constraints by polynomials. In our sumcheck protocol, we first commit $(\mathbf{z}, \mathbf{r})$ as a multilinear polynomial over $\mathbb{F}_{q^k}$. Let $\tilde{\mathbf{w}}$ be a multilinear polynomial such that

$$\tilde{\mathbf{w}}(u, \ell) = \begin{cases} z_{u,\ell}, & \text{if } u \leq \mu \\ r_{u-\mu,\ell}, & \text{if } \mu < u \leq \mu + n \end{cases} \quad (21)$$

Here, we slightly abuse the notation; that is, u and ℓ actually refer to the binary representations of numbers in $[\mu + n]$ and $[d]$, respectively. Now we can assume that α is fixed. We define multilinear polynomials $\tilde{\alpha}$ and $\tilde{M}_\alpha$ such that:

$$\forall i, u, \ell, \quad \tilde{\alpha}(\ell) = \alpha^\ell \quad \text{and} \quad \tilde{M}_\alpha(i, u) = \begin{cases} \mathbf{M}_{i,u}(\alpha), & \text{if } u \leq \mu \text{ and } i \leq n \\ -(\alpha^d + 1), & \text{if } i + \mu = u \\ 0, & \text{otherwise} \end{cases}$$

Now, for each fixed α , we can represent the constraints as polynomials:

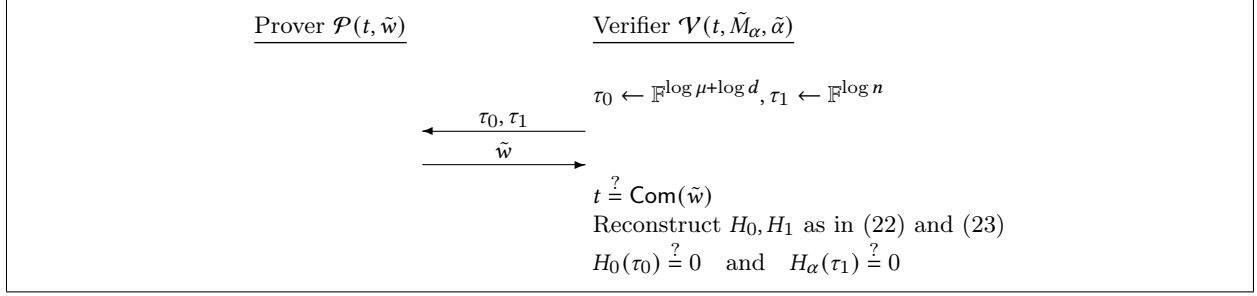


Fig. 5. Proving that polynomials H_0 and H_α are zero via random evaluation.

– For each $i \in [n]$,

$$\sum_{u=1}^{\mu+n} \tilde{M}_\alpha(i, u) \sum_{\ell} \tilde{w}(u, \ell) \cdot \tilde{\alpha}(\ell) = \mathbf{y}_i(\alpha),$$

where $\mathbf{M}_{i,j}, \mathbf{y}_i \in \mathbb{Z}_q^{<d}[X]$.

– For each $u \in [\mu + n]$ and $\ell \in [d]$,

$$\tilde{w}(u, \ell) \cdot (\tilde{w}(u, \ell) - 1) \cdot (\tilde{w}(u, \ell) + 1) \cdots (\tilde{w}(u, \ell) - b + 1) \cdot (\tilde{w}(u, \ell) + b - 1) = 0.$$

Following standard approaches [Set20; HMZ25], we use an equality function to batch these constraints. Since $i \in [n]$, its binary representation is a string in $\{0, 1\}^{\log n}$. Recall that the equality function $\tilde{eq} : \mathbb{F}_{q^k}^{\log n} \times \mathbb{F}_{q^k}^{\log n} \rightarrow \mathbb{F}_{q^k}$ is defined

$$\tilde{eq}(t, i) = \prod_j (t_j \cdot i_j + (1 - t_j) \cdot (1 - i_j))$$

Using $\tilde{eq}$, we can then batch the first type of constraints to show that the following polynomial $H_\alpha(t) : \mathbb{F}_{q^k}^{\log n} \rightarrow \mathbb{F}_{q^k}$ is a zero polynomial.

$$H_\alpha(t) := \sum_{i \in [n]} \tilde{eq}(t, i) \left(\sum_{u, \ell} \tilde{M}_\alpha(i, u) \cdot \tilde{w}(u, \ell) \cdot \tilde{\alpha}(\ell) - \mathbf{y}_i(\alpha) \right), \quad (22)$$

Similarly, we can also batch the second type of constraints (smallness check) as proving that the following polynomial $H_0 : \mathbb{F}_{q^k}^{\log \mu + \log d} \rightarrow \mathbb{F}_{q^k}$ is a zero polynomial.

$$H_0(t) := \sum_{t, u} \tilde{eq}(t, (u, \ell)) \cdot \tilde{w}(u, \ell) (\tilde{w}(u, \ell) - 1) (\tilde{w}(u, \ell) + 1) \cdots (\tilde{w}(u, \ell) - b + 1) (\tilde{w}(u, \ell) + b - 1). \quad (23)$$

Finish the proof using sumcheck protocols. Now, the goal of the honest prover is to prove that $H_0(t) \equiv 0$ and $H_\alpha(t) \equiv 0$. We adopt a standard approach again: the verifier sends two random points, τ_0 and τ_1 , and the prover then proves that $H_0(\tau_0) = 0$ and $H_\alpha(\tau_1) = 0$. We capture this with the following protocol. Similarly as in Lemma 9, we can show coordinate-wise special soundness of this protocol by treating (τ_0, τ_1) as a vector of $\log \mu + \log d + \log n$ coordinates.

Lemma 10 (CWSS of Figure 5). *There is an efficient, deterministic algorithm that given a statement $(t, \tilde{M}_\alpha, \tilde{\alpha})$ and $D := \max(2d, 2b - 1)$ valid transcripts $((\tau_{i,0}, \tau_{i,1}), \tilde{w}_i)$ of the protocol in Figure 5 such that $(\tau_{i,0}, \tau_{i,1})_i \in \text{SS}(\mathbb{F}_{q^k}, 2, D)$, can either extract a valid opening $\tilde{w}$ of t , i.e. $t = \text{Com}(\tilde{w})$, which satisfies $H_0(t_0) \equiv 0$ and $H_\alpha(t) \equiv 0$ (defined in Equations (22) and (23)), or break binding of the commitment scheme Com .*

Proof. If for some $i \neq j$ we have $\tilde{w}_i \neq \tilde{w}_j$, then we break the binding property of the commitment scheme. Otherwise, define $\tilde{w} := \tilde{w}_i$ for all i . By definition of $D = \max(2d, 2b - 1)$ and the coordinate-wise special soundness, we have found at least $2d$ (resp. $2b - 1$) distinct roots for H_α (H_0), which implies that the aforementioned polynomials are equal to zero. $\square$

Now, fix τ_0, τ_1 and recall that by definition of H_0 and H_α :

$$H_0(\tau_0) := \sum_{\ell, u} \tilde{e}q(\tau_0, (u, \ell)) \tilde{w}(u, \ell) (\tilde{w}(u, \ell) - 1) (\tilde{w}(u, \ell) + 1) \cdots (\tilde{w}(u, \ell) - b + 1) (\tilde{w}(u, \ell) + b - 1),$$

and

$$H_\alpha(\tau_1) = \sum_{u, \ell} \tilde{w}(u, \ell) \cdot \tilde{\alpha}(\ell) \sum_i \tilde{e}q(\tau_1, i) \cdot \tilde{M}_\alpha(i, u) - \sum_i \tilde{e}q(\tau_1, i) \cdot \mathbf{y}_i(\alpha).$$

We then use sumcheck to prove that $H_0(\tau_0) = H_\alpha(\tau_1) = 0$. Defining polynomials

$$F_{0, \tau_0}(x, y) := \tilde{e}q(\tau_0, (x, y)) \tilde{w}(x, y) (\tilde{w}(x, y) - 1) (\tilde{w}(x, y) + 1) \cdots (\tilde{w}(x, y) - b + 1) \cdot (\tilde{w}(x, y) + b - 1) \cdot 1_{\leq \mu}(x, y),$$

and

$$F_{\alpha, \tau_1}(x, y) := \tilde{w}(x, y) \cdot \tilde{\alpha}(y) \cdot \sum_i \tilde{e}q(\tau_1, i) \cdot \tilde{M}_\alpha(i, x),$$

we have that $H_0(\tau_0) = 0$ and $H_\alpha(\tau_1) = 0$ if and only if

$$\sum_{u, \ell} F_{0, \tau_0}(u, \ell) = 0 \quad \text{and} \quad \sum_{u, \ell} F_{\alpha, \tau_1}(u, \ell) = a,$$

where $a := \sum_i \tilde{e}q(\tau_1, i) \cdot \mathbf{y}_i(\alpha)$, which the verifier can compute on their own. Therefore, we lastly apply sumcheck on F_{0, τ_0} and F_{α, τ_1} .

A reinterpretation of the sumcheck protocol. In the standard sumcheck protocol, the prover and the verifier communicate univariate polynomials and random field elements. In the last round of the protocol, the verifier makes a single evaluation query to the polynomial. However, since we rely on the special soundness notion (see Lemma 11), we reinterpret the sumcheck protocol as a composition of single-round communication protocols to simplify the analysis. In Figure 6, we demonstrate the sumcheck protocol for polynomials of the form $H(\mathbf{X}) := P(\mathbf{X}) \cdot Q(\tilde{\mathbf{w}}(\mathbf{X}))$, where P and Q are public information and $\tilde{\mathbf{w}}(\mathbf{X})$ is the witness. We note that both F_{0, τ_0} and F_{α, τ_1} have this form.

Compared to the standard sumcheck protocol, we note that the prover usually does not send $\tilde{\mathbf{w}}$ to the verifier directly. Instead, it typically proves knowledge of $\tilde{\mathbf{w}}$ by continuing the sumcheck protocol, and finally by providing a proof of the evaluation of $\tilde{\mathbf{w}}$.

Lemma 11 (Special soundness of Figure 6). *There is an efficient, deterministic algorithm that given a statement and $D := \deg(H) + 1$ valid transcripts $(g_i(X_i), a_{i,j}, \tilde{w})_j$ of the protocol in Figure 6 such that $(a_{i,j})_j \in \text{SS}(\mathbb{F}_{q^k}, 1, D)$, can either extract a valid opening $\tilde{w}$ of t , i.e. $t = \text{Com}(\tilde{w})$, which satisfies $\sum_{b_j} H(a_1, \dots, a_i, b_{i+1}, \dots, b_\ell) = g_i(a_i)$, or break binding of the commitment scheme Com .*

Proof. If for some $i \neq j$ we have $\tilde{w}_i \neq \tilde{w}_j$, then we break the binding property of the commitment scheme. Otherwise, define $\tilde{w} := \tilde{w}_i$ for all i . By definition of $D = \deg(H) + 1$ and the coordinate-wise special soundness, we have found at least $\deg(H) + 1$ distinct roots for $\sum_{b_j} H(a_{<i}, X_i, b_{i+1}, \dots, b_\ell) - g_i(X_i)$, which implies that $\sum_{b_j} H(a_{<i}, X_i, b_{i+1}, \dots, b_\ell) - g_i(X_i) \equiv 0$.

Since special soundness is composable, we can obtain the special soundness of the sumcheck protocol by composing Figure 6 for all $i \in [\ell]$. Finally, the prover outputs $y' := \tilde{w}(a_1, \dots, a_\ell)$, which allows the verifier to compute the evaluation $H(a_1, \dots, a_\ell)$. What is left for the prover to do now is prove the polynomial evaluation claim for $\tilde{w}$. This ends the first iteration of our protocol.

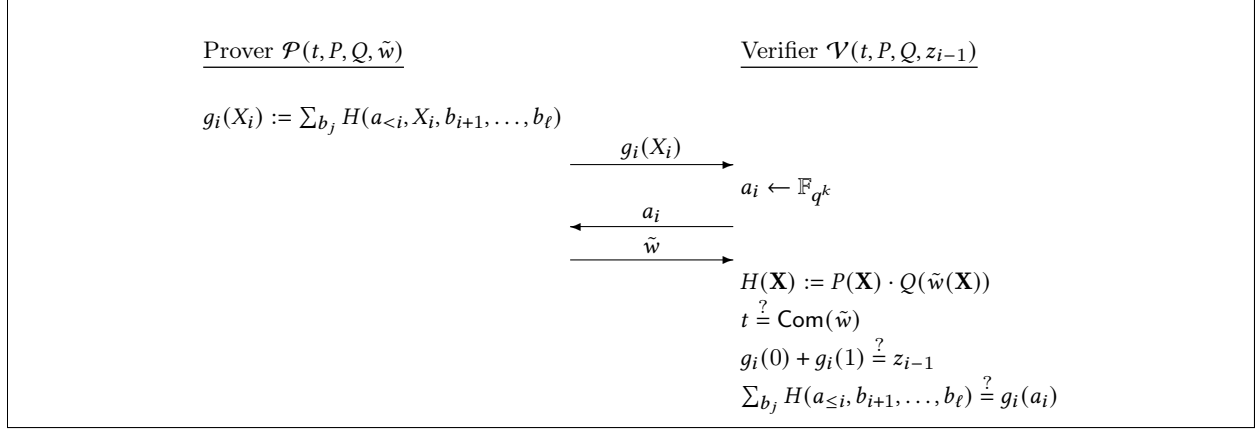


Fig. 6. Proving $\sum_{b_i, \dots, b_\ell} H(a_1, \dots, a_{i-1}, b_i, \dots, b_\ell) = z_{i-1}$ for $i \in [\ell]$, where $H(\mathbf{X}) := P(\mathbf{X}) \cdot Q(\tilde{w}(\mathbf{X}))$. We set z_0 to be the initial sum, and $z_i := g_i(a_i)$.

Putting everything together. The complete protocol is presented in Figure 7, which is a natural composition of subprotocols in Figures 4, 5 and 6. By combining Lemmas 9, 10 and 11, we conclude that our final protocol satisfies coordinate-wise special soundness.

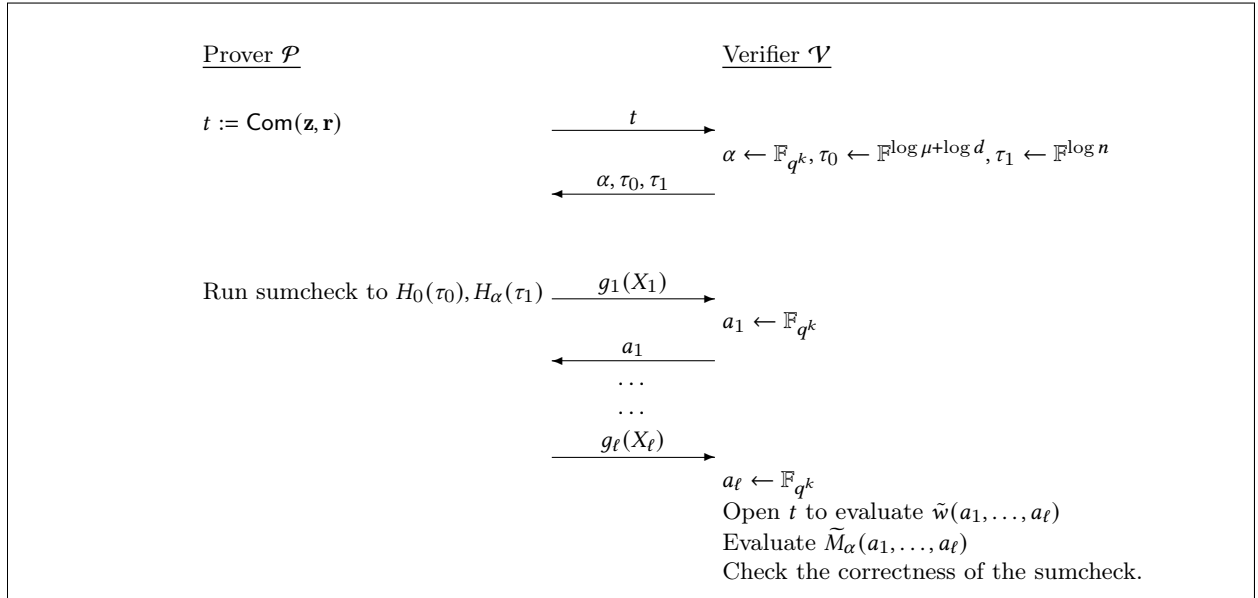


Fig. 7. Efficient Sumcheck via Ring Switching

4.4 Asymptotic Efficiency Analysis

Suppose we want to prove knowledge of the committed multilinear polynomial $f \in \mathbb{Z}_q^{\leq 1}[X_1, \dots, X_\ell]$ which satisfies a polynomial evaluation claim over $\mathbb{F}_{q^k}$. In the following, let $L := 2^\ell$.

Basic parameters. Let $q = \text{poly}(\lambda, L)$ and $k = \lambda/\log q$, so that $\mathbb{F}_{q^k}$ is of exponential size to ensure negligible knowledge soundness in Lemmas 9, 10 and 11. We set the ring dimension $d = \text{poly}(\lambda)$, so $\alpha = \log \lambda$, and $n_A = n_B = n_D = O(1)$ for hardness of the Module-SIS problem. Then, we pick $\omega = O(d)$ for sufficiently large challenge space so that knowledge soundness error in Lemma 8 becomes negligible. We pick the decomposition base $b = O(1)$ and therefore $\delta = O(\log q)$. Recall that β in Section 4.2 was the upper-bound on the vector $\mathbf{z}$ (see Figure 3). A naive bound for β is:

$$\left\| \sum_{i=1}^{2^r} c_i \mathbf{s}_i \right\|_{\infty} \leq \sum_{i=1}^{2^r} \omega \cdot b = 2^r \omega b =: \beta.$$

Hence, if we choose the smallest integer τ such that $b^\tau > \beta$, then $\tau = O(r + \alpha)$.

Witness length reduction after one iteration. By Section 3, we can consider polynomials over $\mathbf{R}_q$ with $\ell - \alpha$ variables. As in Section 4.2, we define variables m, r such that $m + r = \ell - \alpha$. Let $m = r = (\ell - \alpha)/2$ (we implicitly assume $\ell - \alpha$ is even, which does not have any *asymptotic* impact). Then, the witness $(\hat{\mathbf{t}}, \hat{\mathbf{w}}, \hat{\mathbf{z}})$, parsed in Section 4.3 as $\mathbf{z}$, together with the “quotient” $\mathbf{r}$ from the ring switching methodology, is together of length – in terms of $\mathbb{Z}_q$ elements:

$$\underbrace{(n_A + 1) \cdot 2^r \cdot \delta \cdot d}_{|\hat{\mathbf{t}}| + |\hat{\mathbf{w}}|} + \underbrace{(\tau + 1) 2^m \cdot \delta \cdot d}_{|\hat{\mathbf{z}}|} + \underbrace{(n_A + n_B + n_D + 2) \cdot \delta \cdot d}_{|\mathbf{r}|} = O((\ell + \alpha)^2 \cdot 2^{\frac{\ell + \alpha}{2}}).$$

Hence, we conclude that after one round of iteration, the number of variables for the polynomial evaluation gets reduced (at most) to:

$$\ell \mapsto \frac{\ell + \alpha}{2} + 2 \log(\ell + \alpha) + O(1) \leq \frac{2}{3}(\ell + \alpha) + O(1).$$

After t iterations, the length of the witness is reduced to (at most):

$$\left(\frac{2}{3}\right)^t \cdot \ell + \sum_{i=1}^t \left(\frac{2}{3}\right)^i \cdot \alpha + O(t) \leq \left(\frac{2}{3}\right)^t \cdot \ell + 2\alpha + O(t).$$

Thus, after running our protocol $t = O(\log \ell)$ times we will output a witness polynomial with at most $O(\log \ell + \alpha)$ variables. Consequently, sending all the $\mathbb{Z}_q$ -coefficients of the final polynomial in the clear costs $\text{poly}(\ell, d) = \text{poly}(\ell, \lambda)$ bits.

Proof size. Having non-interactive proofs via Fiat-Shamir transformation in mind, we only count the prover messages. The proof size of a single iteration can be split in three parts. The first is the transformation from $\mathbb{Z}_q$ to $\mathbf{R}_q$ from Section 3, where the prover sends $d + k(k - 1)$ $\mathbb{Z}_q$ -elements, where $k = \lambda/\log \lambda$. The second is the commitment $\mathbf{v}$ from Section 4.2, which is of size $n_D \cdot d$ elements in $\mathbb{Z}_q$. Finally, in the protocol in Section 4.3, for each sumcheck round (out of $O(\ell + \alpha)$) the prover sends 2 (resp. $b + 1$) elements in $\mathbb{F}_{q^k}$ corresponding to the coefficients of the univariate polynomials obtained from F_{α, τ_1} (resp. F_{0, τ_0}). Finally, the prover sends the commitment to $\tilde{\mathbf{w}}$ which is of length $n_B \cdot d$ $\mathbb{Z}_q$ -elements (resp. Section 4). In total, a single iteration of the protocol has proof size, in terms of $\mathbb{Z}_q$ elements:

$$(n_B + n_D + 1) \cdot d + (k - 1) \cdot k + (b + 2) \cdot k \cdot O(\ell + \alpha) = O\left(\frac{\lambda}{\log \lambda} \cdot \ell\right) + \text{poly}(\lambda).$$

By recursively running the protocol $O(\log \ell)$ times and by having $q = \text{poly}(\lambda, L)$, we obtain succinct $\text{poly}(\ell, \lambda)$ proof size.

Verifier time. We split the verifier algorithm in three parts. For the transformation in Section 3 the verifier has to compute the $\mathbf{R}_q$ -trace map consistency, which is of time $\text{poly}(\lambda)$. For the protocol in Section 4.2 the verifier does not check anything. Finally, in Section 4.3 the verifier has two tasks. First is checking that the univariate polynomials sent by the prover every sumcheck round are consistent. This takes $O(b) = O(1)$ $\mathbb{Z}_q$ -operations. A much more expensive part is the final evaluation of the polynomial F_{α, τ_1} after sumcheck, which involves evaluating the (publicly known) multilinear extensions $\tilde{M}_\alpha$ $n := n_B + n_D + 3 = O(1)$ times. As shown in [VSBW13], to evaluate $\tilde{M}_\alpha$ once, it would take

$$O((n_A + 1)2^r + (\tau + 1)2^m \delta) = \tilde{O}(\sqrt{2^\ell / \lambda}) \text{ time}$$

in terms of field operations over $\mathbb{F}_{q^k}$ using dynamic programming. A single multiplication over $\mathbb{F}_{q^k}$ can be computed with $\tilde{O}(k)$ operations over $\mathbb{Z}_q$. Since $k = O(\lambda / \log q)$ the total verifier time is dominated by $\tilde{O}(\sqrt{2^\ell \cdot \lambda})$.

4.5 Concrete Optimizations and Composing with Greyhound

Avoiding re-decomposition. Having a recursive proof system in mind, a naive approach to commit to the multilinear extension $\text{mle}[\tilde{w}]$ of $\tilde{w}$ (cf. Equation (21)) is to run again the commitment scheme from Section 4. The first part is the inner commitment, where we decompose the coefficients of $\tilde{w}$ (and thus the committed witness length increases), and then run an Ajtai commitment scheme to get the inner commitments $(\mathbf{t}_i)_i$. A more natural and efficient way is to directly commit to $\tilde{w}$ (cf. Figure 7), as we know that the coefficients of $\tilde{w}$ are already short and no decomposition for $\tilde{w}$. It is straightforward from the committing point of view, as we show below, but more importantly, how should one prove an evaluation claim over $\mathbb{F}_{q^k}$:

$$\text{mle}[\tilde{w}](a_1, \dots, a_\ell) = y,$$

when having only committed to $\tilde{w}$?

Let us recall the notation for completeness. The vector $\tilde{w} \in \mathbb{Z}^{n \cdot d}$ is defined in Equation (21), and in particular, the coefficients of $\tilde{w}$ are between $-\beta$ and β , where parameters $\mathbf{n}$ and β are chosen according to the protocol in Section 4.3. We assume that $\mathbf{n}d$ is a power-of-two. Let $d' = 2^{\alpha'}$ be a power-of-two ring dimension used to define $\mathbf{R}'_q := \mathbb{Z}_q[X]/(X^{d'} + 1)$ (potentially $d' \neq d$); it is the cyclotomic ring used for the next iteration of the protocol. Let m', r' be integers chosen such that $\mathbf{n}d = 2^\ell$, where $\ell = m' + r' + \alpha'$; they will be responsible for the split-and-fold in the next round.

We start with the definition of a multilinear polynomial. Let us define $\mathbf{a}_0 = (a_1, \dots, a_{\ell-\kappa})$ and $\mathbf{a}_1 := (a_{\ell-\kappa+1}, \dots, a_\ell)$. Then, we can write

$$y = \text{mle}[\tilde{w}](a_1, \dots, a_\ell) = \sum_{i \in \{0,1\}^\kappa} eq(i, \mathbf{a}_1) \cdot \underbrace{\left(\sum_{j \in \{0,1\}^{\ell-\kappa}} \tilde{w}_{j||i} \cdot eq(j, \mathbf{a}_0) \right)}_{y_i :=} \quad (24)$$

Hence, we let the prover send k partial evaluations $y_i \in \mathbb{F}_{q^k}$ to the verifier who can directly check¹⁰ whether

$$y = \sum_i eq(i, \mathbf{a}_1) \cdot y_i.$$

What we have left to prove is that all the y_i are well-formed, or in other words $\sum_{j \in \{0,1\}^{\ell-\kappa}} \tilde{w}_{j||i} \cdot eq(j, \mathbf{a}_0) = y_i$ for $i \in \{0,1\}^\kappa$.

¹⁰ As in Section 3, the prover can just send the $k-1$ evaluations y_i excluding $y_{0\dots 0}$, which can be derived directly by the verifier.

For the next part, we assume without loss of generality that $\mathbb{F}_{q^k} = \mathbf{R}'_q{}^H$ where $H = \langle \sigma_{-1}, \sigma_{4k+1} \rangle$ (cf. Section 3). Indeed, as the field isomorphism fixes elements in $\mathbb{F}_q$, the coefficients of $\mathbf{z}$ are still small after mapping the statement into $\mathbf{R}'_q{}^H$. Then, any field element $v \in \mathbf{R}'_q{}^H$ can be explicitly written as:

$$v := v_0 \cdot Z_0 + \sum_{j=1}^{k-1} v_{k-j} \cdot Z_{k-j} \quad \text{where} \quad Z_0 = 1 \text{ and } Z_{k-j} := X^{\frac{d'}{2k} \cdot (k-j)} - X^{\frac{d'}{2k} \cdot (k+j)}.$$

Now, for each $j \in \{0, 1\}^{\ell-\kappa}$ define

$$\hat{w}_j := \sum_{i \in \{0,1\}^\kappa} \tilde{w}_{j\|i} \cdot Z_{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \in \mathbb{F}_{q^k}. \quad (25)$$

We observe that $\|\hat{w}_j\|_\infty \leq \beta$. Then, proving well-formedness of all $(y_i)_i$ is equivalent to proving

$$\begin{aligned} \sum_{j \in \{0,1\}^{\ell-\kappa}} \hat{w}_j \cdot eq(j, \mathbf{a}_1) &= \sum_{j \in \{0,1\}^{\ell-\kappa}} \sum_{i \in \{0,1\}^\kappa} \tilde{w}_{j\|i} \cdot eq(j, \mathbf{x}_1) \cdot Z_{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \\ &= \sum_{i \in \{0,1\}^\kappa} \sum_{j \in \{0,1\}^{\ell-\kappa}} (\tilde{w}_{j\|i} \cdot eq(j, \mathbf{x}_1)) \cdot Z_{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \\ &= \sum_{i \in \{0,1\}^\kappa} y_i \cdot Z_{\sum_{t=1}^\kappa i_t \cdot 2^{t-1}} \in \mathbb{F}_{q^k}. \end{aligned} \quad (26)$$

where the right-hand side is public. Next, we prove (26) as follows. Note that ultimately we can write

$$\sum_{j \in \{0,1\}^{\ell-\kappa}} \hat{w}_j \cdot eq(j, \mathbf{a}_1) = \mathbf{e}^\top (\mathbf{f}^\top \otimes \mathbf{I}_{2^r}) \hat{w}_j \quad (27)$$

where $\mathbf{e} := (eq(i, \mathbf{x}_0))_i \in \mathbb{F}_{q^k}^{2^{2r'}}$ and $\mathbf{f} := (eq(j, \mathbf{x}_1))_j \in \mathbb{F}_{q^k}^{2^{m'} d'/k}$. Finally, using Theorem 2, the $\mathbf{R}_q$ -trace of the ring element

$$p := \mathbf{e}^\top (\sigma_{-1}(\psi(\mathbf{f}))^\top \otimes \mathbf{I}_{2^r}) \psi(\hat{w}) \in \mathbf{R}'_q$$

is exactly equal to the value in (27), which is also equal to (26). By Lemma 6, the infinity norm of $\psi(\hat{w}) \in \mathbf{R}'_q{}^{n \cdot d/d'}$ is at most 2β , so we will commit to it using the inner-outer commitments without initial decomposition. What is left to prove is the knowledge of the short commitment opening $\psi(\hat{w})$ which satisfies the quadratic relation above. This is exactly the type of statement supported natively by Greyhound (cf. [NS24, Section 3]). To sum up, the total communication complexity of our reduction to Greyhound is

$$\underbrace{(k-1) \cdot k \cdot \log q + d' \cdot \log q}_{|y_i|} \quad \underbrace{}_{|p|}. \quad (28)$$

Using Johnson-Lindenstrauss instead of sumcheck. Similarly as in Greyhound [NS24], instead of running the protocol from Section 4.3, at some point we can naturally switch to LaBRADOR [BS23]. This can be done when the witness length is small enough so that performing the Johnson-Lindenstrauss projection becomes cheap. In fact, although the sumcheck-based approach provides a faster verification time, reducing witness length after a single operation is slower. The reason is that we have to pick relatively small base b , as the sumcheck proof size consists of $O(b)$ field elements. This implies longer decomposition δ , and thus longer witness for the next round. On the other hand, the random projection approach allows larger bases b (at a fairly small cost), which makes δ concretely smaller. This allows LaBRADOR to recurse further until the final witness is cheap enough to send in the clear ($\approx 30\text{KB}$).

5 Experimental Results

To demonstrate the concrete performance of our protocol, we have developed a prototype implementation of Hachi in Rust. The source can be found here:

<https://github.com/georgeourke/hachi-pcs>

Our implementation currently supports committing to, and single evaluation proofs and verification for multilinear polynomials over $\mathbb{Z}_q$. We highlight that our implementation is still in an early stage of development and is amenable to a number of optimizations which would lead to considerable increases in performance. Despite this, our implementation already demonstrates significant improvements in verification time, with first (by far most expensive) round verification times being 12 to 29 times faster than Greyhound verification. Additionally, we concretely improve commitment times for large polynomials (30+ variables). At present, our evaluation proof times substantially exceed those of existing implementations and will need further optimization — such as SIMD acceleration — to align with the state of the art.

We have benchmarked our implementation on a University server (with AVX-512 support) and a Mac Mini M4 (no AVX support). The stated times for Prove and Verify are for the first round only; though, following a single round, the new witness is far smaller and so the first round of the protocol dominates the total cost of the proof and verification times.

Hardware	Variables	Commit / s	Prove / s	Verify / ms
Mac Mini M4	30	149	268	96.5
Mac Mini M4	28	34.2	114	49.3
Mac Mini M4	26	8.79	51.3	26.6
Server	30	87.3	495	185
Server	28	21.9	219	93.7
Server	26	5.49	103	39.0
Greyhound	30	132	41.2	2800
Greyhound	28	21.2	8.21	1150
Greyhound	26	4.37	2.03	492

Fig. 8. Concrete running time of the Hachi PCS with prototype implementation on different hardware. Prove and Verify times are for the first (and by far the most computationally expensive) round only. The Greyhound benchmark was tested on an Intel Xeon Sapphire Rapids core at 3.2 GHz with full AVX-512 optimization.

5.1 Concrete Parameters

The parameters (30 variable polynomial) used in benchmarking are shown in Figure 9. We heuristically set the ring dimension $d = 2^\alpha$ of $\mathbf{R}_q$ and the height of the commitment matrices, n , so that $n \cdot d \geq 2^{10}$. For $l = 30$ variables, we aim to reduce the size of the witness as much as possible in ‘one-shot’. Hence, we set $d = 1024$ and $n = 1$. With these parameters, in a single round of Prove and Verify, we reduce the size of the witness by a factor of 128. Specifically, we reduce a 30 variable multilinear polynomial over $\mathbb{Z}_q$ (32 bits per coefficient) to a 26 variable polynomial over $\mathbb{Z}_{16}$ (4 bits per coefficient). The evaluation proof is reduced from an evaluation of a 30 variable polynomial over $\mathbb{Z}_q$ to an evaluation of a 26 variable polynomial with coefficients in $\mathbb{Z}_{16}$ and an evaluation point over F_{q^4} .

5.2 Concrete Efficiency and Composing with Greyhound

To showcase the communication efficiency, let us set parameters for $\ell = 30$. We refer to Section 5 (and in particular Figure 9) for more rationale on the parameters. We aim for $\lambda = 128$ security level and use the Lattice Estimator [APS15] to measure hardness of the Module-SIS problem in the infinity norm.

variable	description	value
ℓ	number of variables of witness polynomial	30
q	prime modulus	4294967197
k	degree of the extension field $\mathbb{F}_{q^k}$ for sumcheck	4
α	log of ring dimension	10
d	$d := 2^\alpha$ is the ring dimension of $\mathbf{R}_q$	1024
m	folding parameter	10
r	folding parameter	10
n_A	height of the matrix $\mathbf{A}$	1
n_B	height of the matrix $\mathbf{B}$	1
n_D	height of the matrix $\mathbf{D}$	1
z	maximum L_∞ norm of $\mathbf{z}$	30583
b	decomposition base	16
δ	expansion factor for decomposing base witness	8
τ	expansion factor for decomposing $\mathbf{z}$	4
ω	L_1 norm of a challenge element	16
c	number of non-zero coefficients in challenge	16
Next witness size		2^{26}
Next witness norm bound	infinity norm	8

Fig. 9. Concrete parameters for an $\ell = 30$ variable multilinear polynomial over $\mathbb{Z}_q$.

To recall, we pick $q \approx 2^{32}$, $d = 2^{10}$, $\alpha = 10$, $r = 10$ and $m = 10$ and $(n_A, n_B, n_D) = (1, 1, 1)$. The decomposition base is $b = 16$, thus $\delta = 8$. For comparison with Greyhound, we consider polynomial evaluations over the base field $\mathbb{F}_q$.

First invocation of Hachi. The cost of the transformation in Section 3 is $d \log q \leq 4\text{KB}$ (where $k = 1$). Next is the commitment $\mathbf{v}$ is also of size 4KB. Length of the witness $\tilde{\mathbf{w}}$ for sumcheck is bounded by 2^{26} , as discussed in Section 5.1. Then, we consider extension fields $\mathbb{F}_{q^k}$ for $k = 4$. The cost of sumcheck is then bounded by $26 \cdot 4 \cdot 32 \cdot (16 + 2)$ bits $\approx 7.3\text{KB}$.

Adapting to the Greyhound relation. We apply our transformation from Section 4.5. To this end, we set the ring dimension exactly as in Greyhound, i.e. $d' = 64$. The *short* witness $\psi(\hat{\mathbf{w}})$ for the quadratic relation over $\mathbf{R}'_q$ is of length $2^{26}/d' = 2^{20}$. By Equation (28), the cost of the transformation can be upper-bounded by:

$$12 \cdot 32 + 64 \cdot 32 \text{ bits} \approx 0.3\text{KB}.$$

Moreover, we also need to commit to $\psi(\hat{\mathbf{w}})$, which again takes 4.5KB (cf. [NS24, Table 4]). Hence, the cost of preparation to run Greyhound is $\approx 4.8\text{KB}$.

Finally, we can directly apply Greyhound for the *short* witness $\psi(\hat{\mathbf{w}})$ of length 2^{20} over $\mathbf{R}'_q$. To estimate the proof size for Greyhound, we consider the parameters of the “ $N = 2^{26}$ ” column of [NS24, Table 1]). This column focuses on the polynomial of length 2^{26} with full-fledged $\mathbb{Z}_q$ -elements. After the coefficient decomposition, the witness gets increased by a factor of $\delta_0 = 5$. Keeping in mind that the ring dimension is $d' = 64$, Greyhound reduces proving polynomial evaluation to a quadratic equation over $\mathbf{R}'_q$, where the witness is of length $2^{26} \cdot \delta/d' \approx 2^{22.3}$, which is more than enough for our scenario. Therefore, we can upper-bound the evaluation proof size of Greyhound is $\approx 43\text{KB}$. To conclude, the total evaluation proof can be estimated to be $7.3 + 4.8 + 43 \text{ KB} = 55.1\text{KB}$.

Running time. The concrete experimental running time for the first round of Hachi can be found in Figure 8. The running time of the transformation, which is done by the prover, from Section 4.5 is negligible compared to other parts. Finally, we use the Greyhound estimates for the witness of length 2^{24} $\mathbb{Z}_q$ -elements, which roughly translates to $2^{24} \cdot \delta/d' \approx 2^{20.3}$ $\mathbf{R}'_q$ -elements. From [ACFY25; Sei24], the prover runtime (which includes

committing to $\psi(\hat{w})$ is around 1220ms, while the verifier runtime is 130ms. Hence, by combining with Figure 8, the total prover and verifier runtime are approximately 270s and 227ms.

5.3 Faster Commitment Time

The commitment function of Hachi is essentially identical to Greyhound except for the choice of ring. Unlike Greyhound, which is restricted to small ring dimensions (e.g. $d = 64$), Hachi is suitable for use with large ring dimensions (e.g. $d = 1024$). A major benefit of this is improved commitment time. This is because the number of ring multiplications decreases in proportion to d^2 (since both 2^{r+m} and n_A, n_B decrease linearly with d) however, with NTT, the cost of the ring multiplication scales with $d \log d$.

The existing Greyhound implementation is highly optimized for certain rings and does not support large ring dimensions. Therefore, we cannot benchmark the commitment times for both the Greyhound parameters and our parameters using the Greyhound implementation. However, as our library is much more flexible, we are able to benchmark the Greyhound commitment times with our implementation - providing a near like-for-like comparison. Our parameters are restricted to powers of two, and we set them as close to the Greyhound parameters as possible. The comparative results are shown in Figure 10 and indicate a 3-5X *algorithmic* improvement in the commitment time.

Variables	Folding Parameters, r, m	Decomposition base	Commit / s	Comparison
30	16384, 1024	256	433	5.0X slower
28	8192, 512	256	71.8	3.3X slower
26	4096, 256	256	18.6	3.4X slower

Fig. 10. Concrete running time of the Commitment function using a smaller ring dimension $d = 64$ on an AVX-512 compatible University server and comparison to concrete times with Hachi parameters.

5.4 Implementation Details

Integer Operations and Data Structures Integer coefficients are stored as unsigned 64 or 32 bit integers (for compatibility with tfhe-ntt). Integer decomposition has been optimized for power of two bases but not optimized with SIMD. Sampling in $\mathbb{Z}_q$ for k -bit prime q is performed by sampling 32/64 bits (using any PRNG) and masking k bits. Rejection sampling is then used to sample uniformly over $\mathbb{Z}_q$. Polynomials are represented as coefficient slices over integers (u32 or u64). Vectors and matrices over polynomials are stored as tightly packed coefficients in a single vector (Array of Structure format). Decomposition of vectors of polynomials is done by sequentially decomposing coefficients and has not yet been optimized with SIMD. Initialization of the commitment matrices is done by sequentially sampling coefficients from a seeded PRNG. If the parameters are set so that the commitment matrices have the same dimensions, the same commitment matrix is used for all commitments. The commitment matrices are re-generated by the commit/prove/verify functions for each invocation and this overhead is included in the stated benchmarks.

Field Arithmetic We use the extension field F_{q^4} to ensure approximately 128 bits of security. Representation and multiplication of field elements in F_{q^4} is provided by the ark-ff library¹¹. In general, we do not use ark-ff for multiplication over the base field $\mathbb{Z}_q$, instead opting to perform integer multiplication over u64 and reduce modulo q naively. This is to maintain compatibility with tfhe-ntt and, whilst suboptimal, avoids conversion between ark-ff types and u64 which is more costly than naive modulo reduction. In several cases we need to multiply an element of the base field $\mathbb{Z}_q$ with an element of the extension field $\mathbb{F}_{q^4}$, and have optimized the arithmetic for this case.

¹¹ <https://crates.io/crates/ark-ff>

Polynomial Arithmetic Hachi requires general polynomial arithmetic over $\mathbf{R}_q := \mathbb{Z}_q[X]/(X^d + 1)$, for a range of ring dimensions d , as well as over $\mathbb{Z}_q[X]$ (to transition to the sumcheck). Our arithmetic library builds on top of the `tfhe-ntt` library¹², which provides SIMD (AVX-2 or AVX-512) optimized NTT-multiplication. Since the prime modulus q cannot be chosen to be ‘NTT-friendly’, we cannot use NTT-based multiplication directly for $\mathbf{R}_q$. Instead, we use the `tfhe-ntt` 64-bit ‘Native NTT’, which performs NTT-based multiplication modulo several primes p_i to simulate the arithmetic modulo $\prod p_i$. Since $\prod p_i$ is larger than the coefficients of any product of polynomials, the product holds over $\mathbb{Z}[X]/(X^d + 1)$ with 64-bit wrapping arithmetic. We then naively reduce the coefficients modulo q to achieve efficient multiplication over $\mathbb{Z}_q[X]/(X^d + 1)$. For the reduction modulo q to hold, we require that no 64-bit wrap has occurred during the multiplication (i.e. the product has coefficients smaller than 64 bits). Note that Hachi only ever requires multiplication over $\mathbf{R}_q$ when one operand is small (i.e. it has been decomposed). With a maximum ring dimension of 2^{10} and 32-bit modulus q , the product will not wrap as long as the coefficients of the small operand are no more than 22 bits. We use a 4-bit decomposition base and so can safely use the 64-bit implementation. The reduction modulo q in our implementation is suboptimal, however it does not add too much to the cost of multiplication (about 5%). On the other hand, whilst this approach lacks optimizations that could be made for more specific rings, it is very flexible; allowing both the (power-of-two) ring dimension and arbitrary modulus q to be chosen at runtime.

To efficiently multiply polynomials over $\mathbb{Z}_q[X]$, note that Hachi only requires computing a sum over a product of *two* polynomials with degree less than d , so that the result has degree at most $2d - 1$. We therefore pad the polynomials to degree $2d - 1$ and use the existing library to efficiently multiply over $\mathbb{Z}_q[X]/(X^{2d} + 1)$. The result does not incur any reduction modulo $X^{2d} + 1$ and so holds over $\mathbb{Z}_q[X]$. Given the full convolution, we perform (optimized) polynomial division to compute the residues modulo $(X^d + 1)$.

Our library abstracts the ring multiplication for larger structures like vectors and matrices of polynomials. We minimize the number of NTTs that must be performed by keeping re-used and intermediate values in NTT form.

Optimized Sparse Polynomial Multiplication Since Hachi may use large ring dimensions, the challenge polynomials can be very sparse whilst still producing an exponential challenge set. For example, for $d = 1024$, the challenge set may consist of ring elements with exactly $k = 16$ non-zero coefficients, each of which is -1 or 1. We sample such elements using a Fisher-Yates-like algorithm to obtain a partial permutation on $0, \dots, d - 1$, giving the indices of the 16 non-zero coefficients. A bit is independently sampled for each non-zero coefficient to determine its sign. Since these challenge elements are so sparse, they can be more efficiently multiplied by a full ring element (compared to treating the sparse element as an arbitrary ring element). Our library implements this sparse polynomial product directly for both $\mathbb{Z}_q[X]$ and $\mathbb{Z}_q[X]/(X^d + 1)$. This has not yet been optimized with SIMD, yet performs about as well as the general AVX-512 optimized NTT multiplication. With a good SIMD optimization, it should significantly outperform general NTT multiplication.

Witness Streams The witness is supplied to the commit/prove functions as a general stream. In our tests, the witness was stored in a file and provided as a file stream which reads chunks of the witness on demand. The commit and prove functions access the witness from the stream in small chunks of size $2^{-m} \approx 0.1\%$ of the whole witness size. This makes our implementation suitable for very large witnesses whilst maintaining moderate memory usage. The overhead of reading the witness from the file stream (as well as re-decomposing in the prove function) is included in the benchmarking.

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¹² <https://crates.io/crates/tfhe-ntt>

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